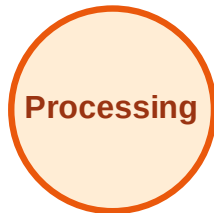


# BFS Traversal

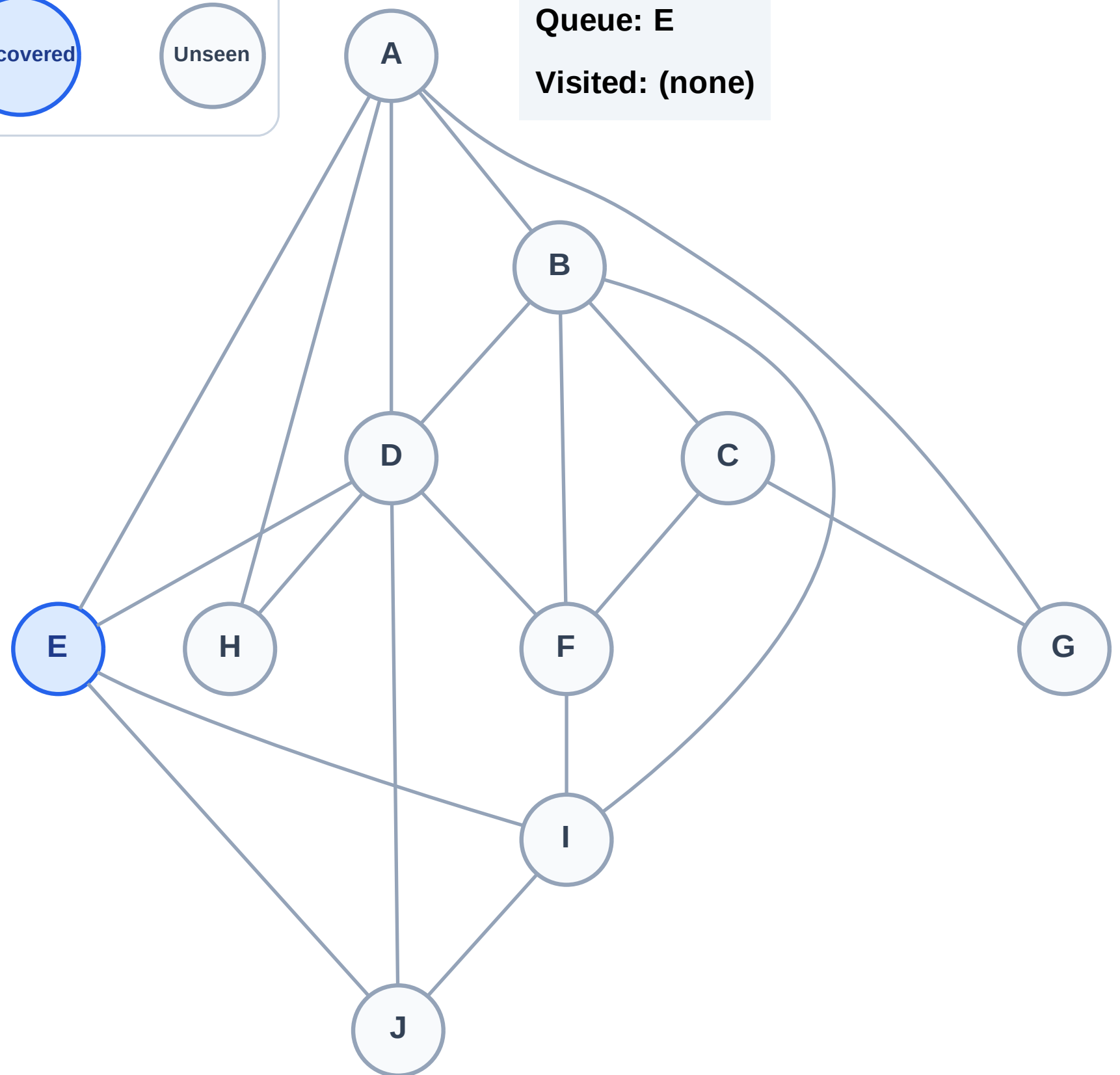
Step 1: Start at E

## Legend



Queue: E

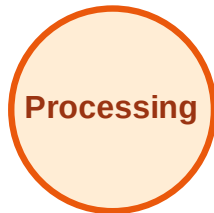
Visited: (none)



# BFS Traversal

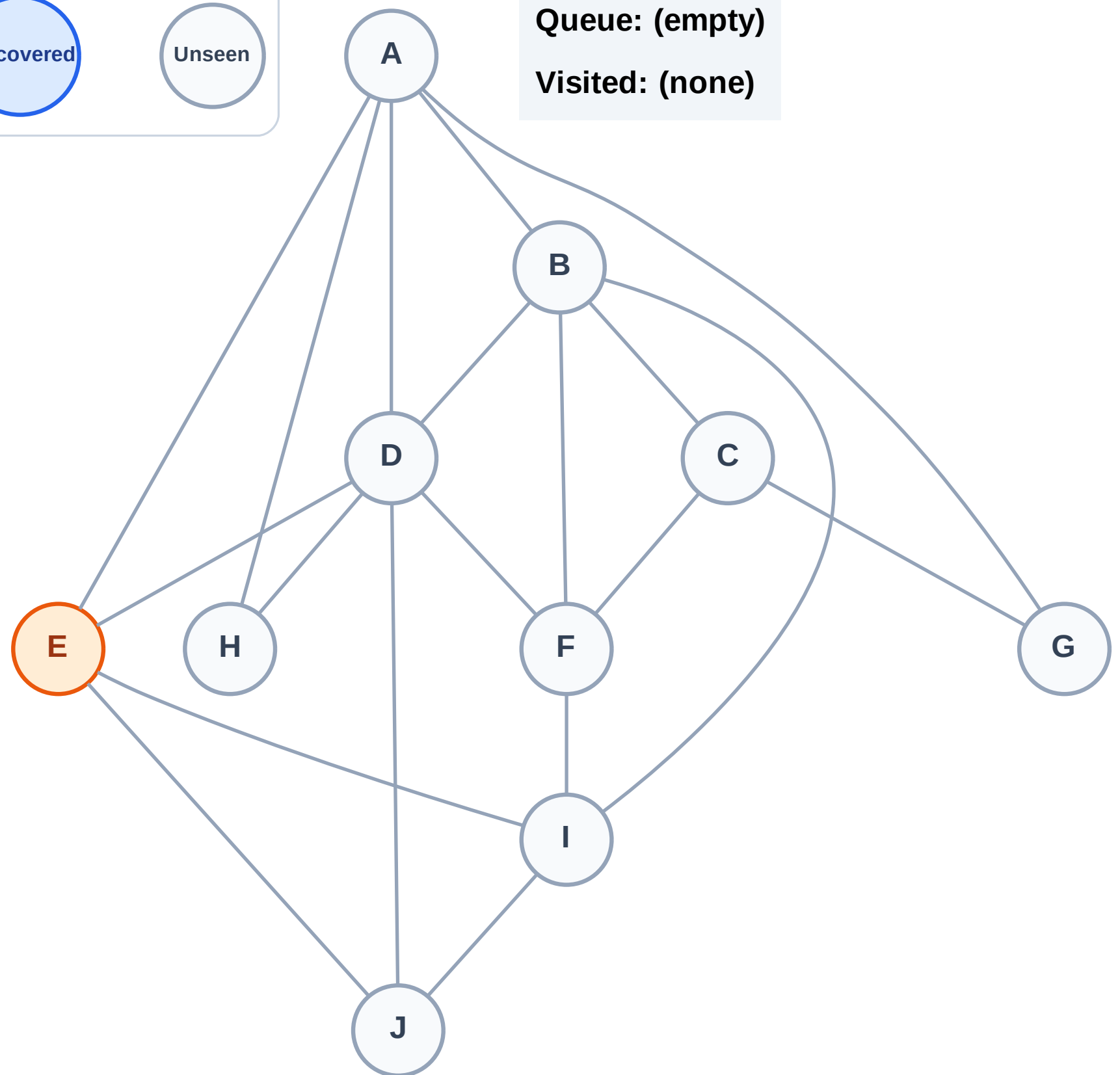
Step 2: Process node E

## Legend



Queue: (empty)

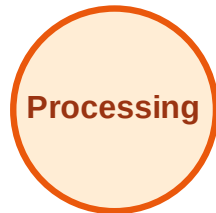
Visited: (none)



# BFS Traversal

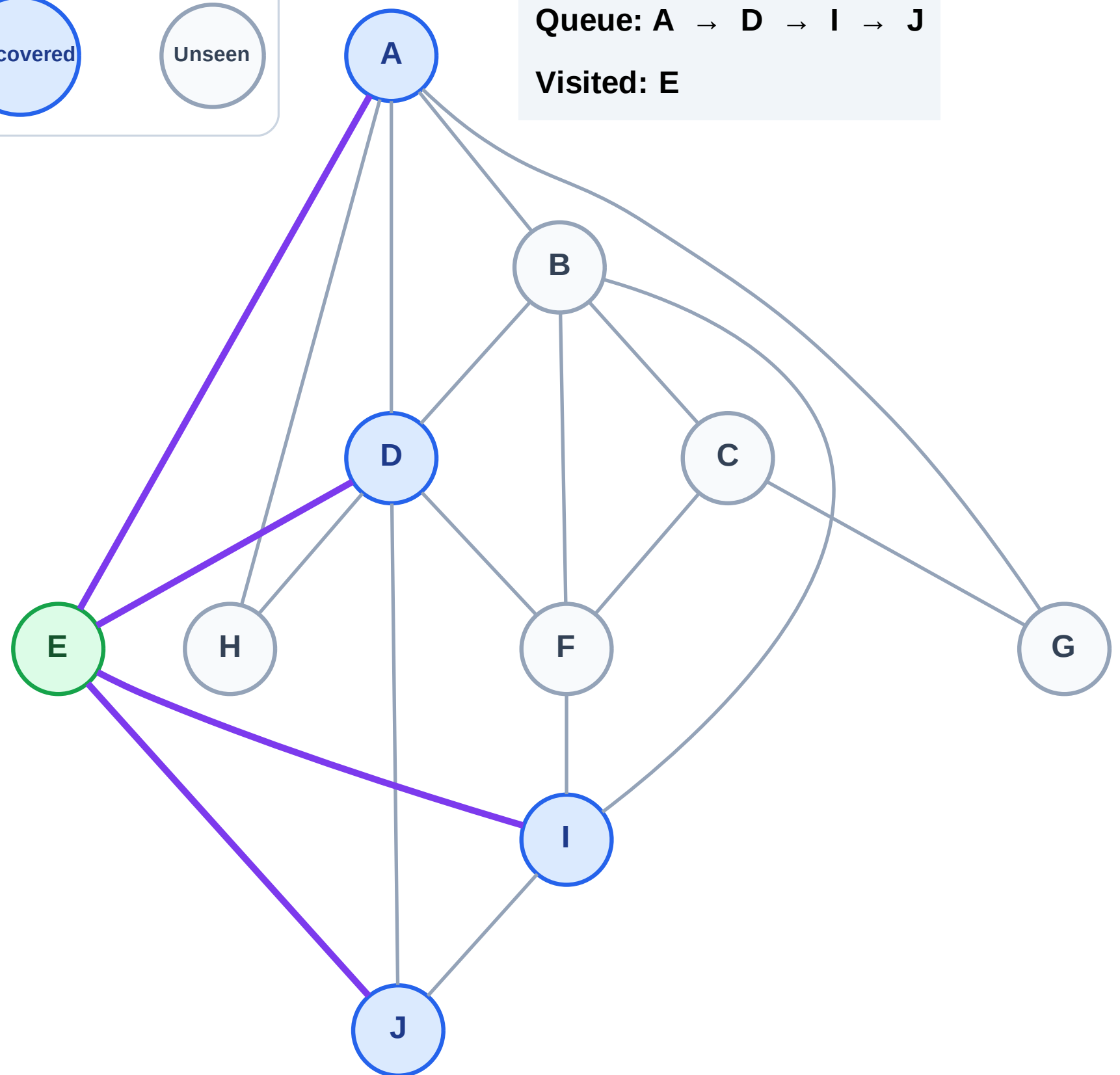
Step 3: Discover A, D, I, J from E

## Legend



Queue: A → D → I → J

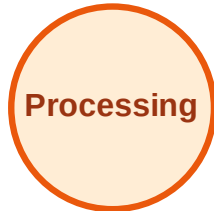
Visited: E



# BFS Traversal

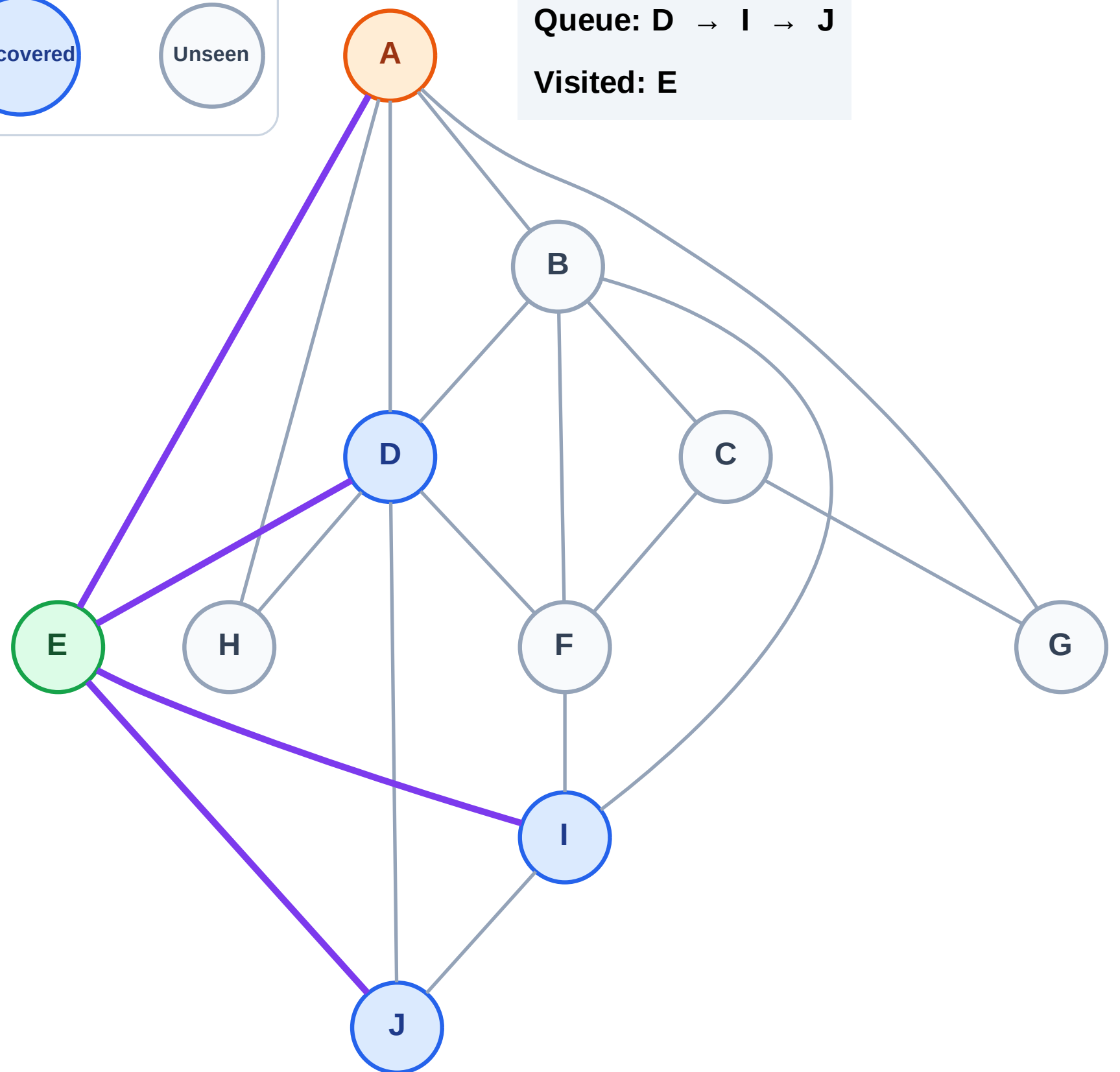
Step 4: Process node A

## Legend



Queue: D → I → J

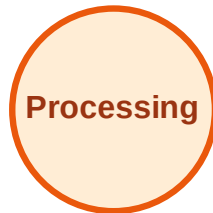
Visited: E



# BFS Traversal

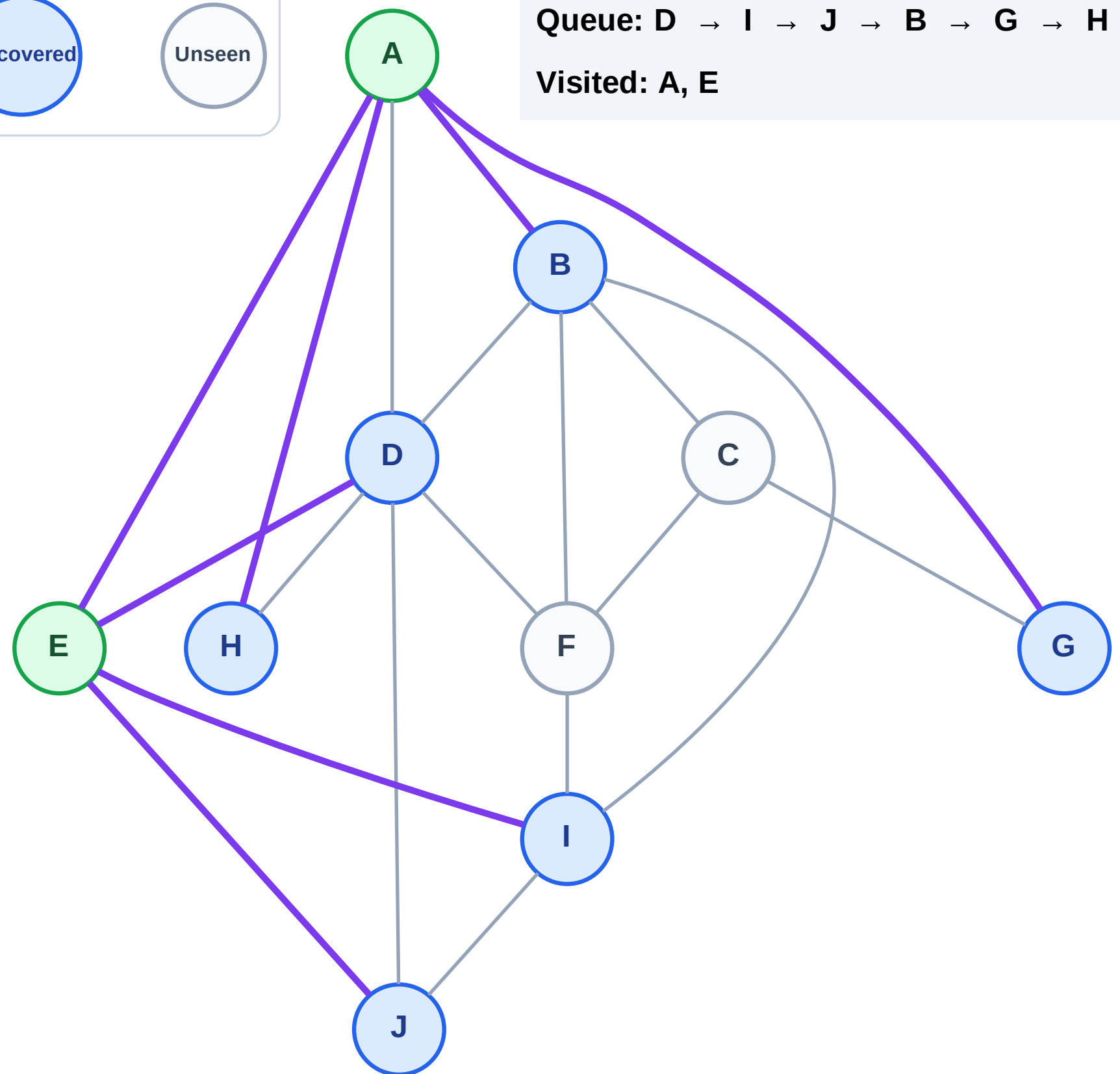
Step 5: Discover B, G, H from A

## Legend



Queue: D → I → J → B → G → H

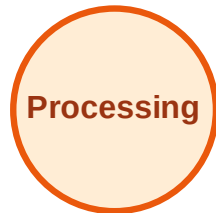
Visited: A, E



# BFS Traversal

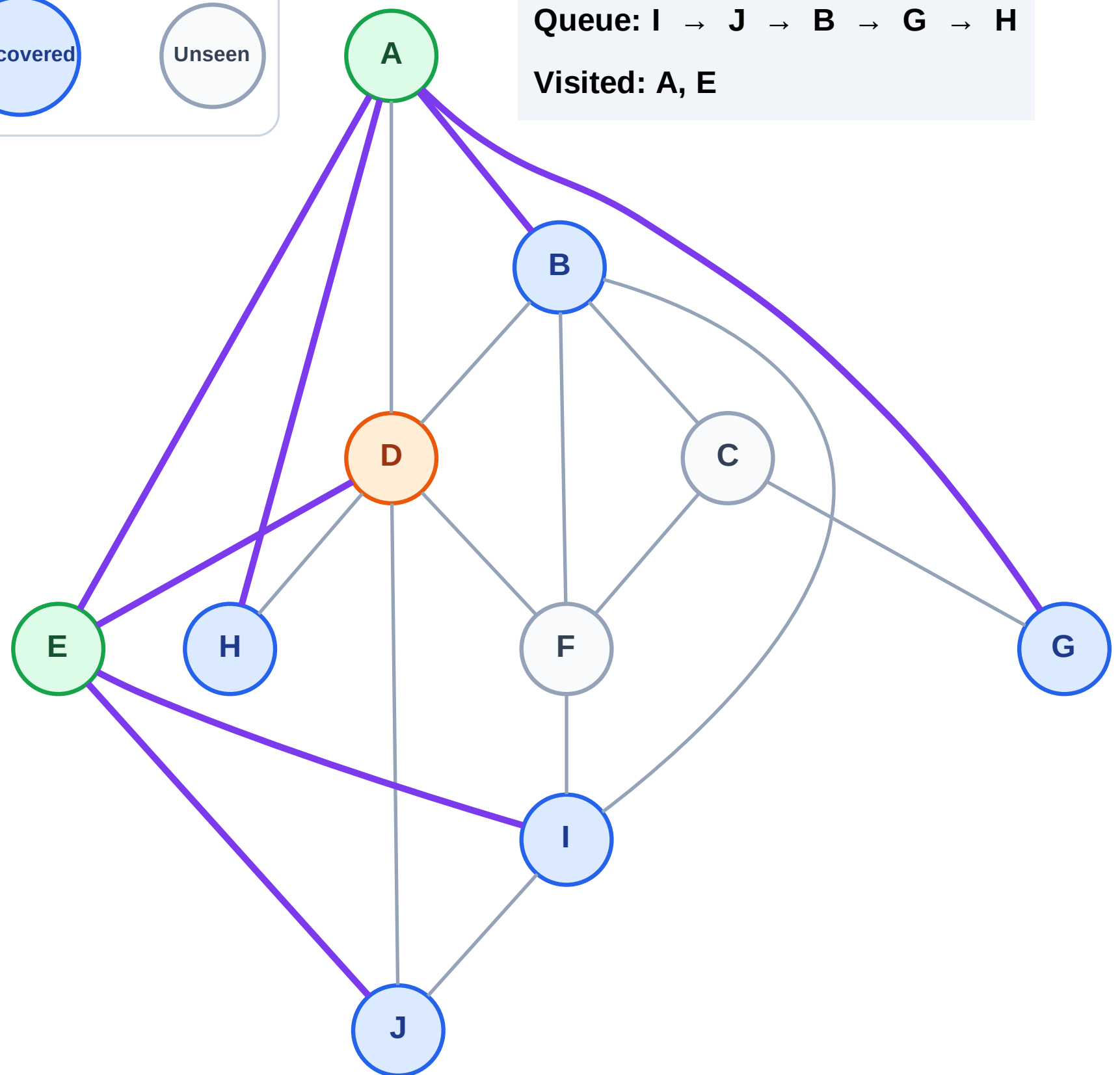
Step 6: Process node D

## Legend



Queue: I → J → B → G → H

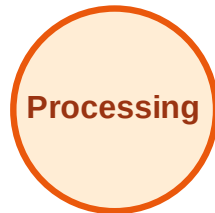
Visited: A, E



# BFS Traversal

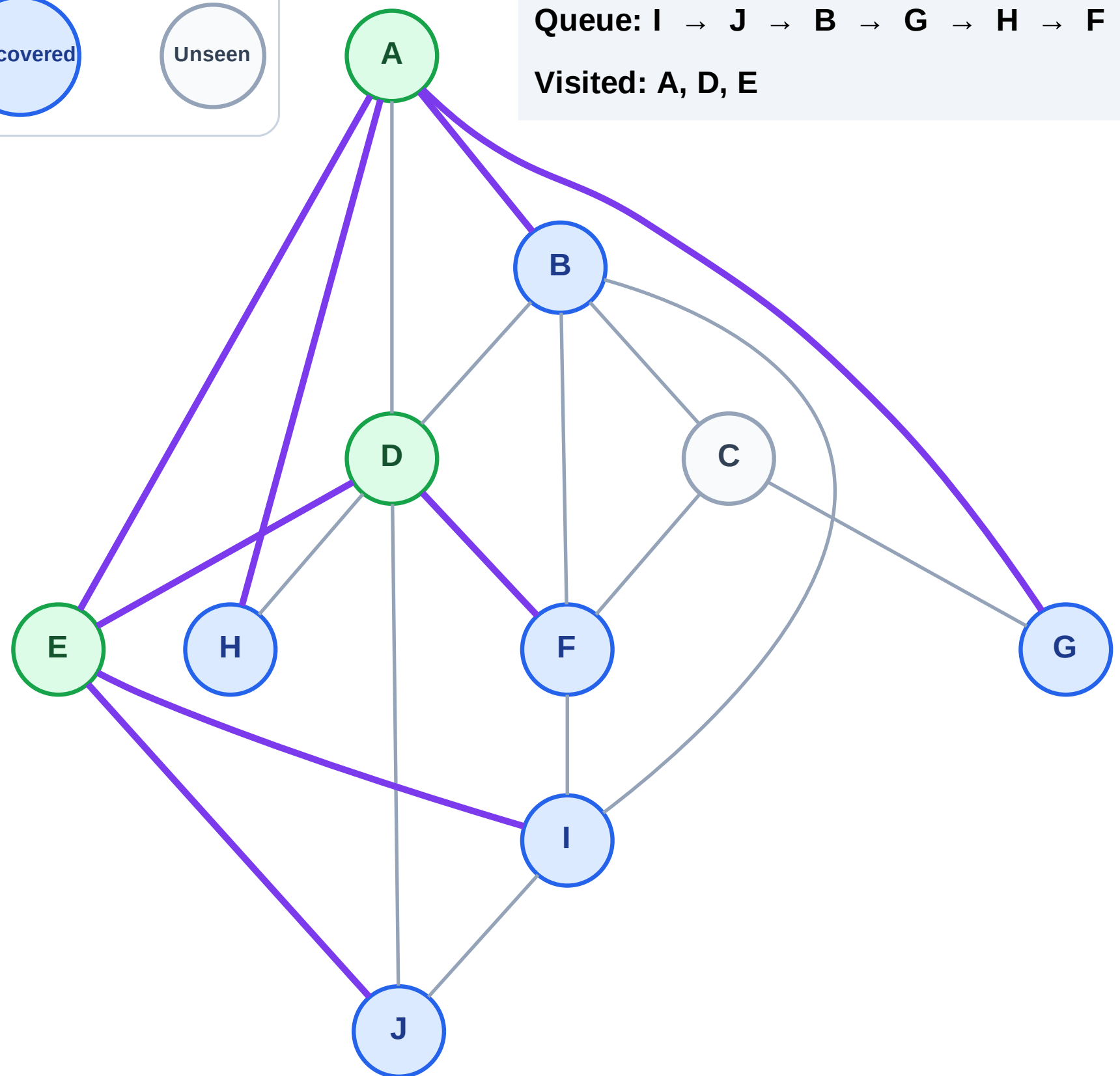
Step 7: Discover F from D

## Legend



Queue: I → J → B → G → H → F

Visited: A, D, E

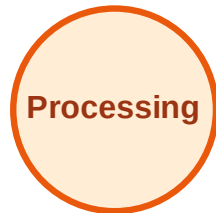




# BFS Traversal

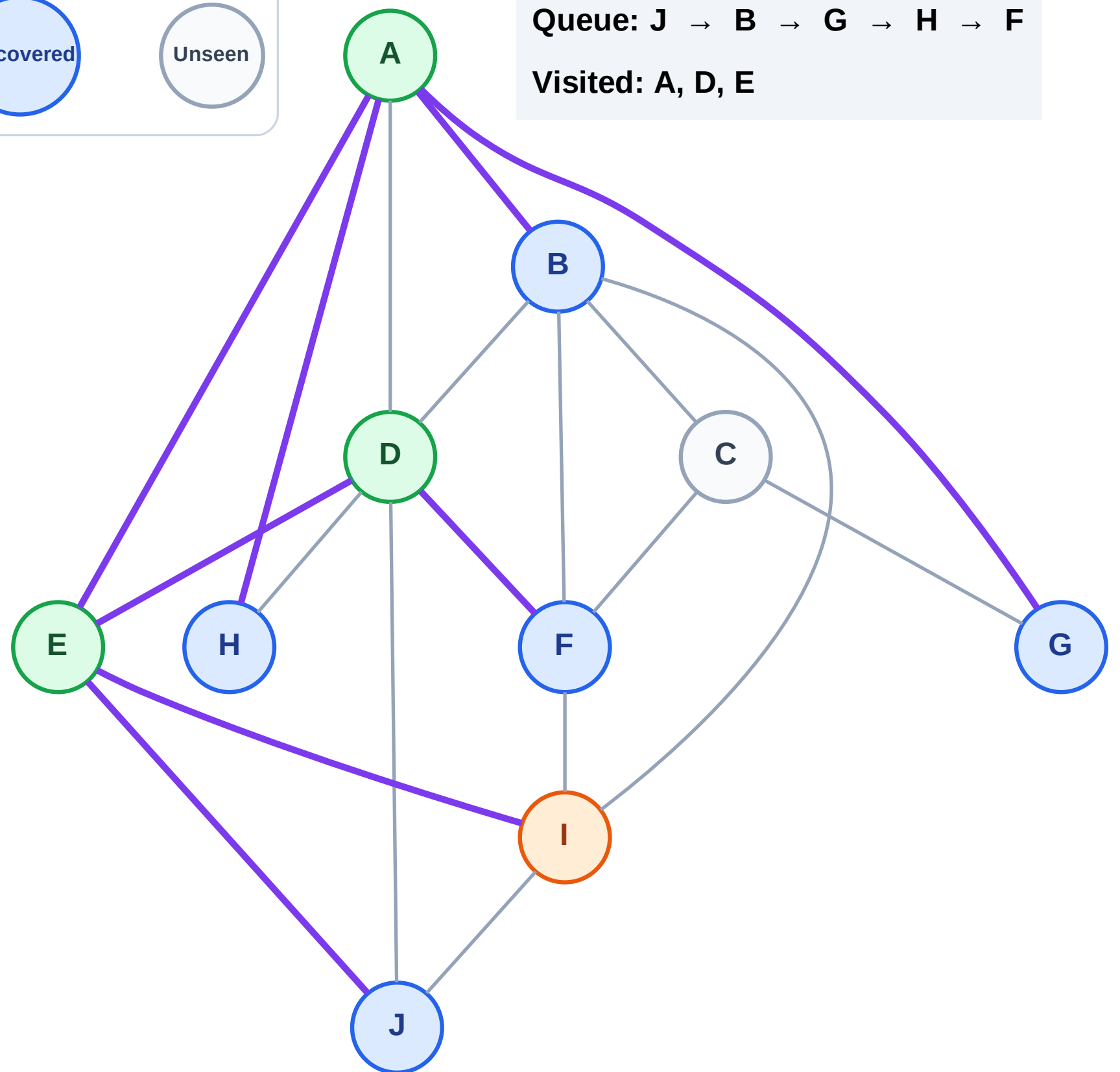
Step 8: Process node I

## Legend



Queue: J → B → G → H → F

Visited: A, D, E

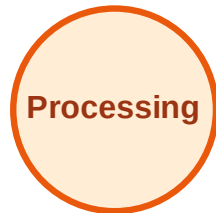




# BFS Traversal

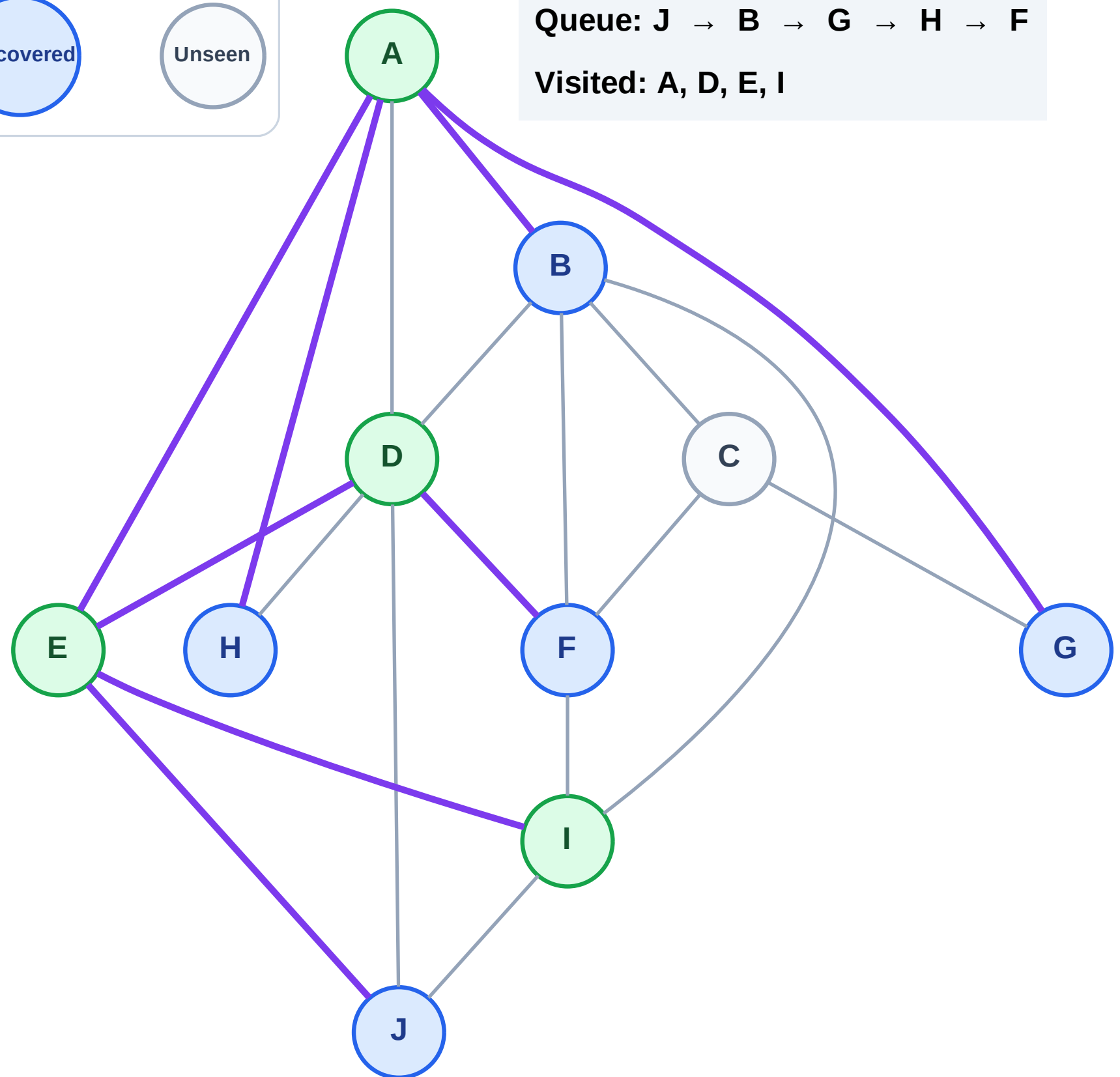
Step 9: No new neighbors from I

## Legend



Queue: J → B → G → H → F

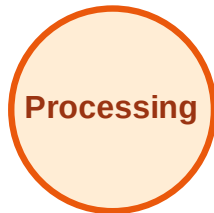
Visited: A, D, E, I



# BFS Traversal

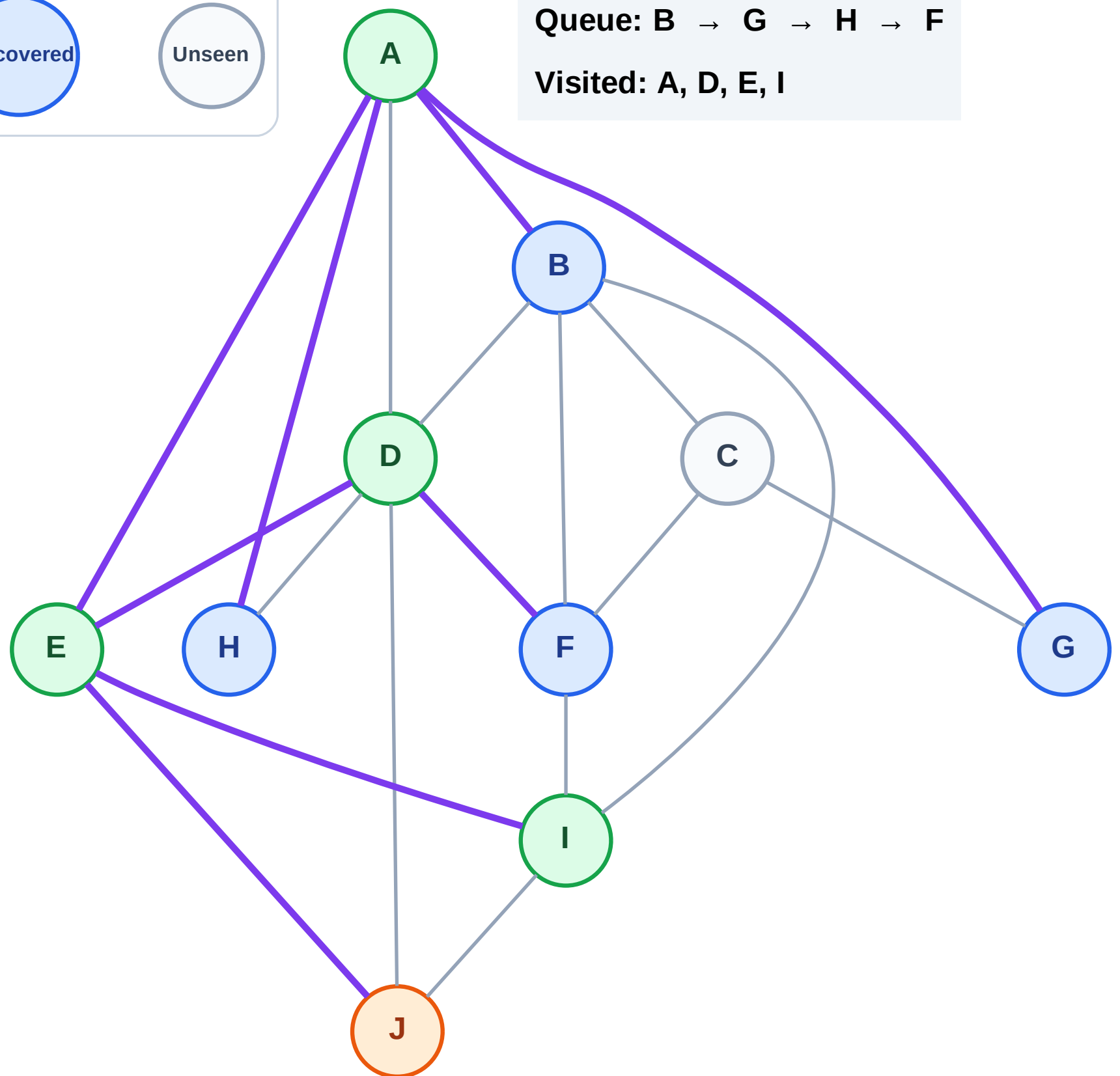
Step 10: Process node J

## Legend



Queue: B → G → H → F

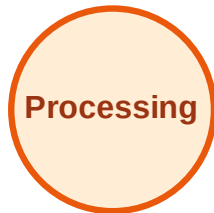
Visited: A, D, E, I



# BFS Traversal

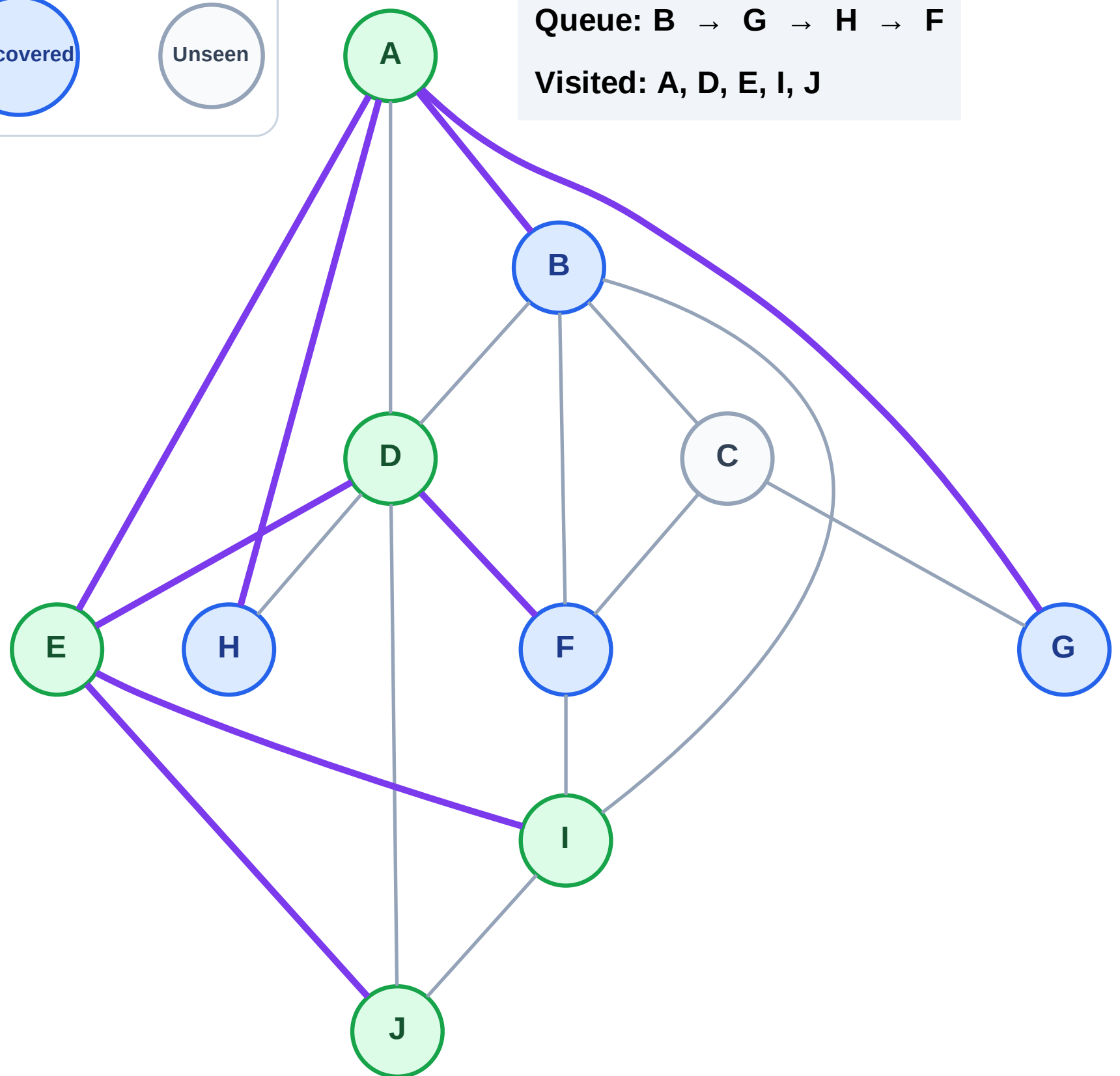
Step 11: No new neighbors from J

## Legend



Queue: B → G → H → F

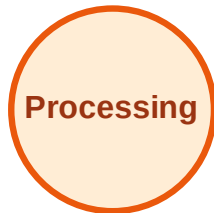
Visited: A, D, E, I, J



# BFS Traversal

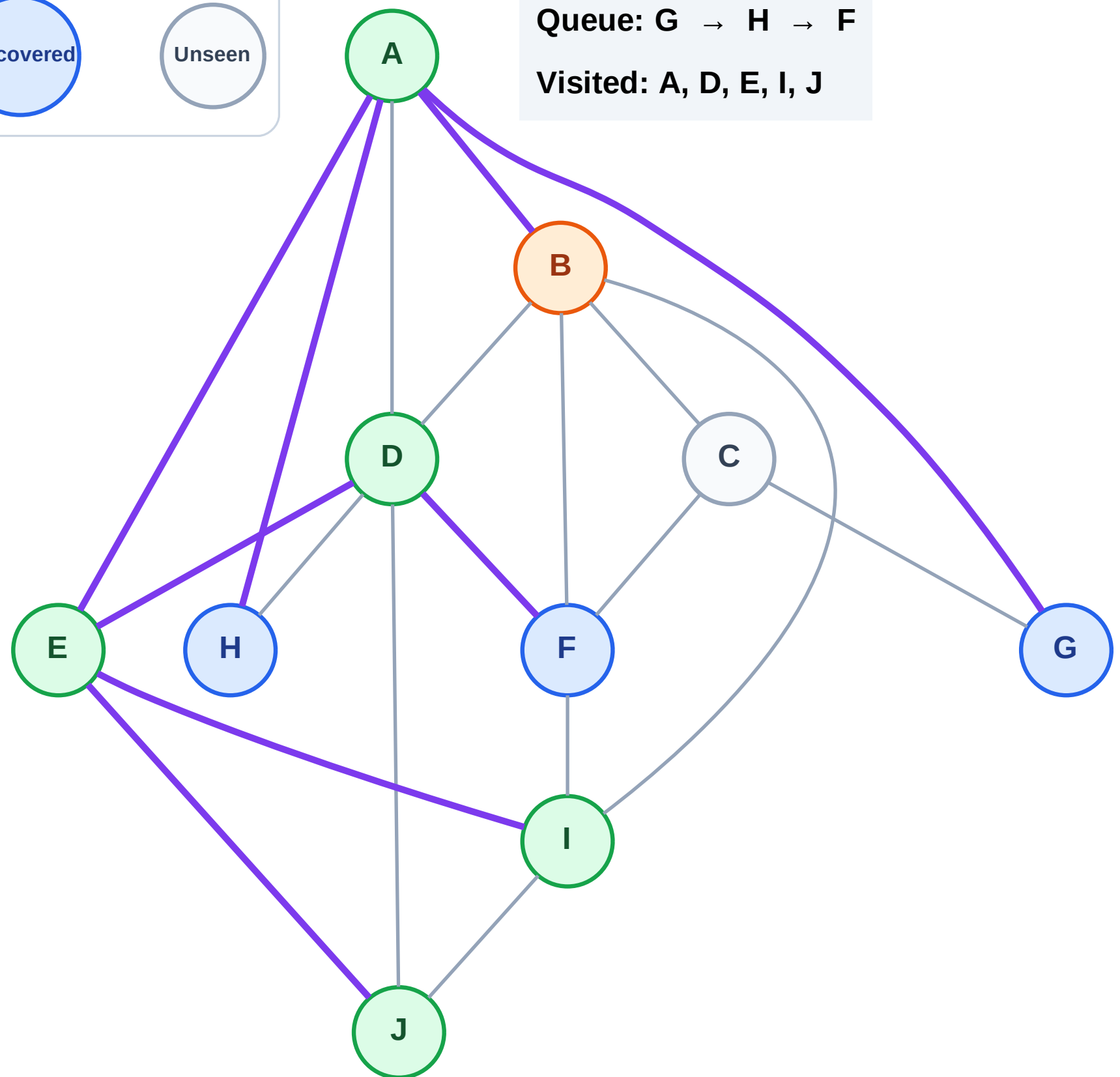
Step 12: Process node B

## Legend



Queue: G → H → F

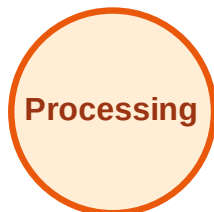
Visited: A, D, E, I, J



# BFS Traversal

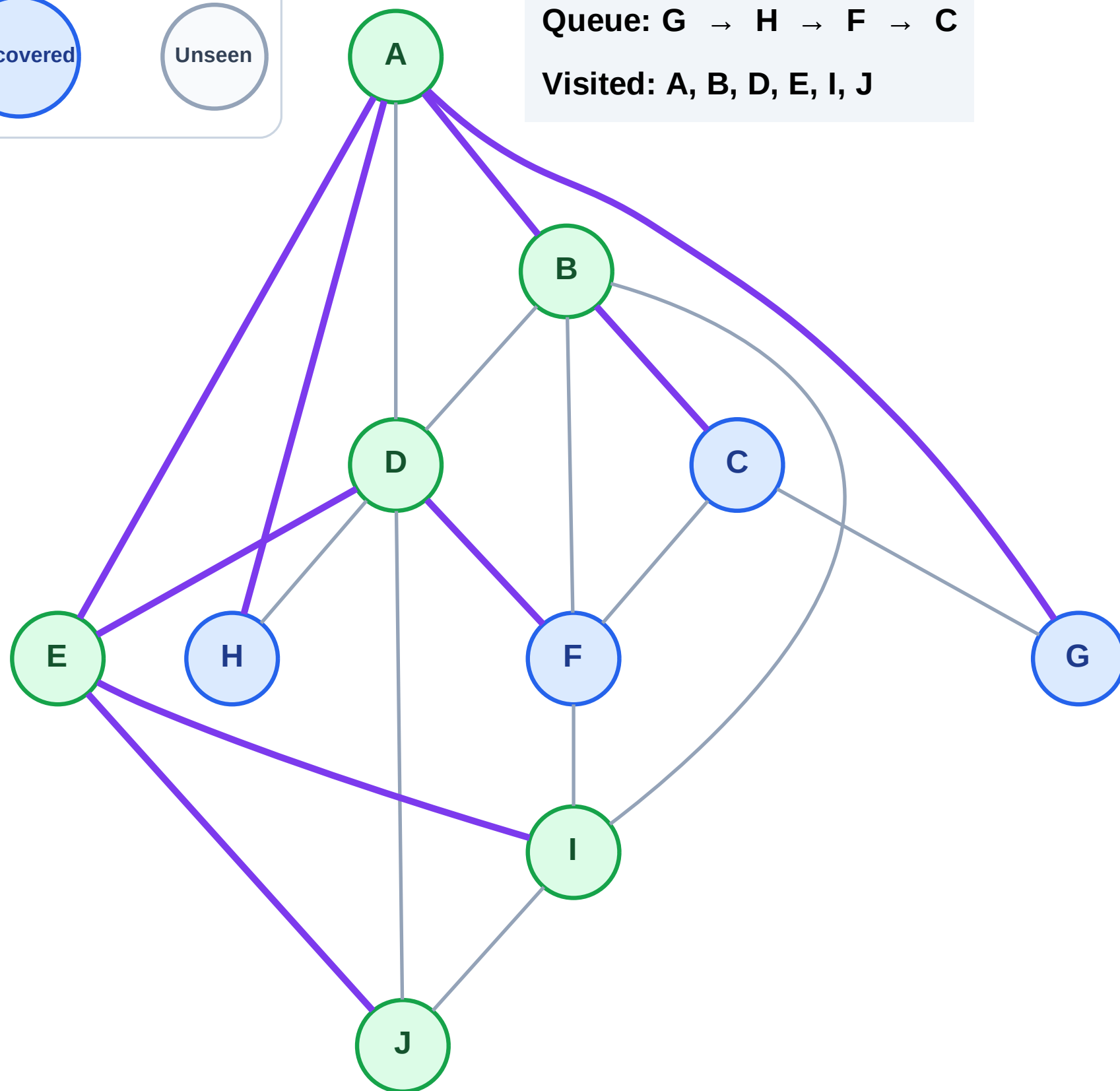
Step 13: Discover C from B

## Legend



Queue: G → H → F → C

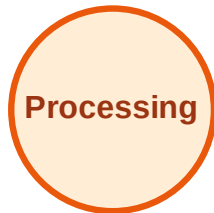
Visited: A, B, D, E, I, J



# BFS Traversal

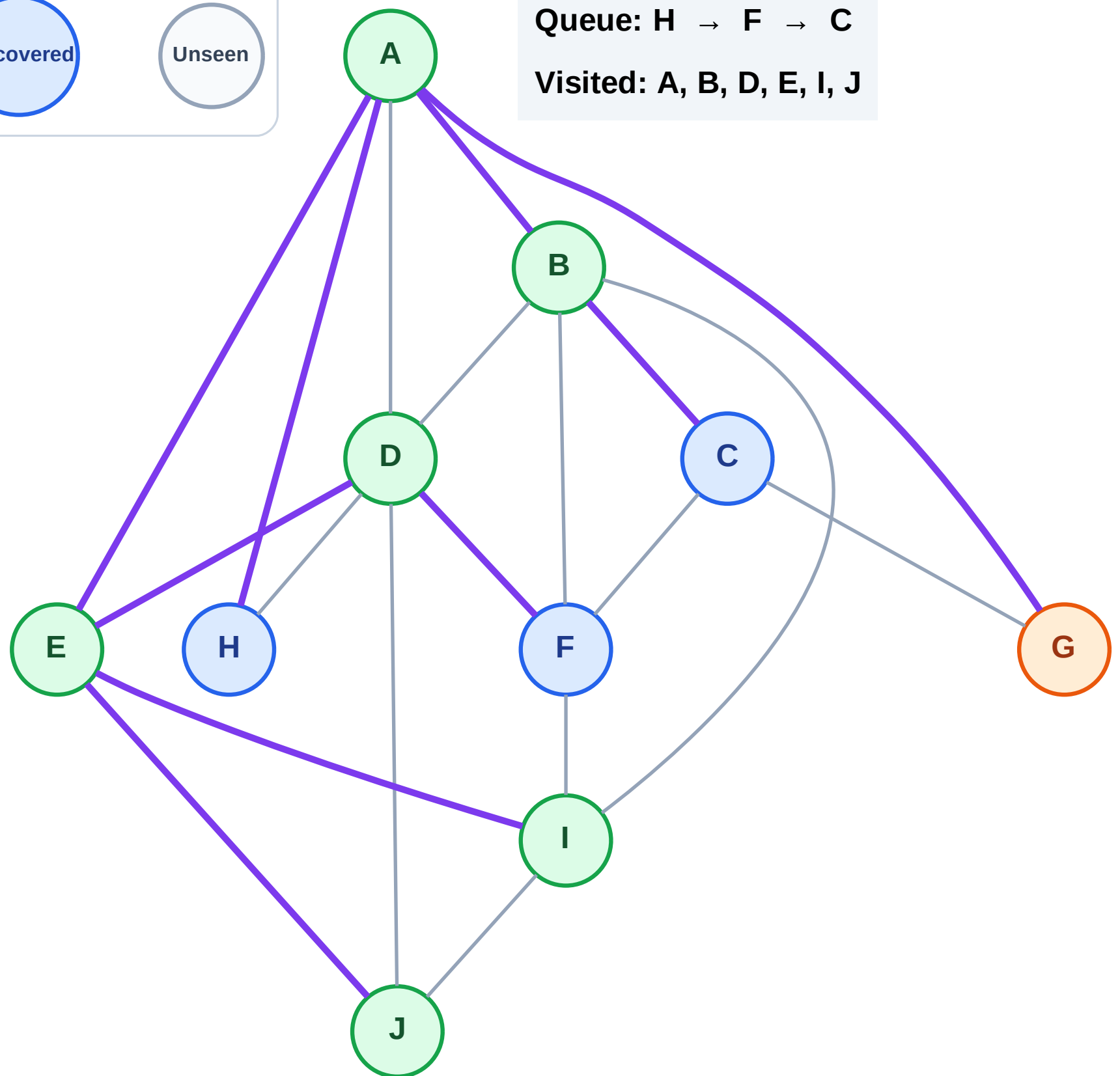
Step 14: Process node G

## Legend



Queue: H → F → C

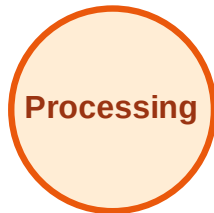
Visited: A, B, D, E, I, J



# BFS Traversal

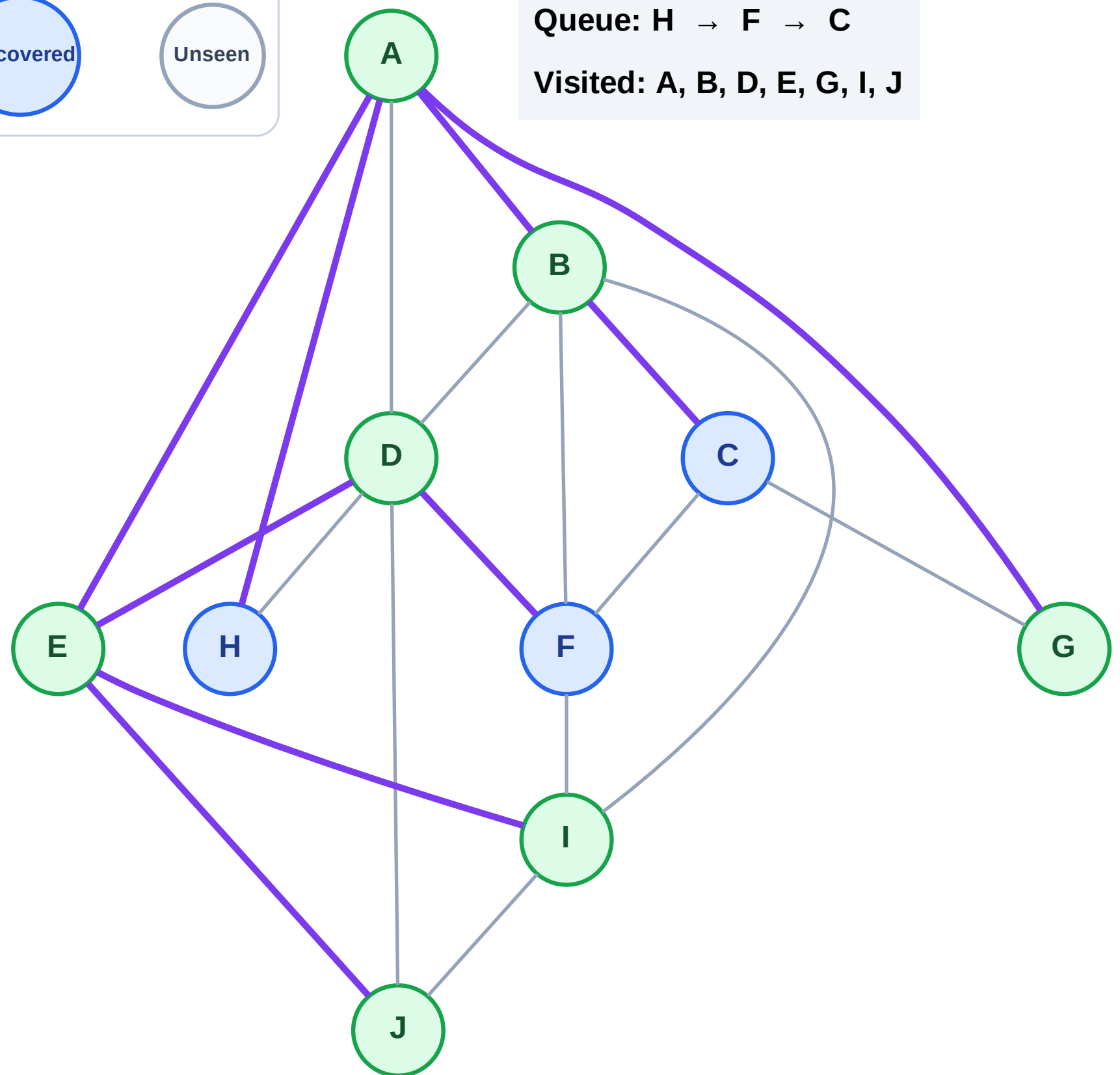
Step 15: No new neighbors from G

## Legend



Queue: H → F → C

Visited: A, B, D, E, G, I, J

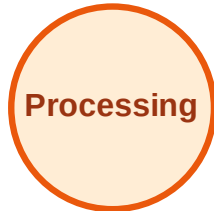




# BFS Traversal

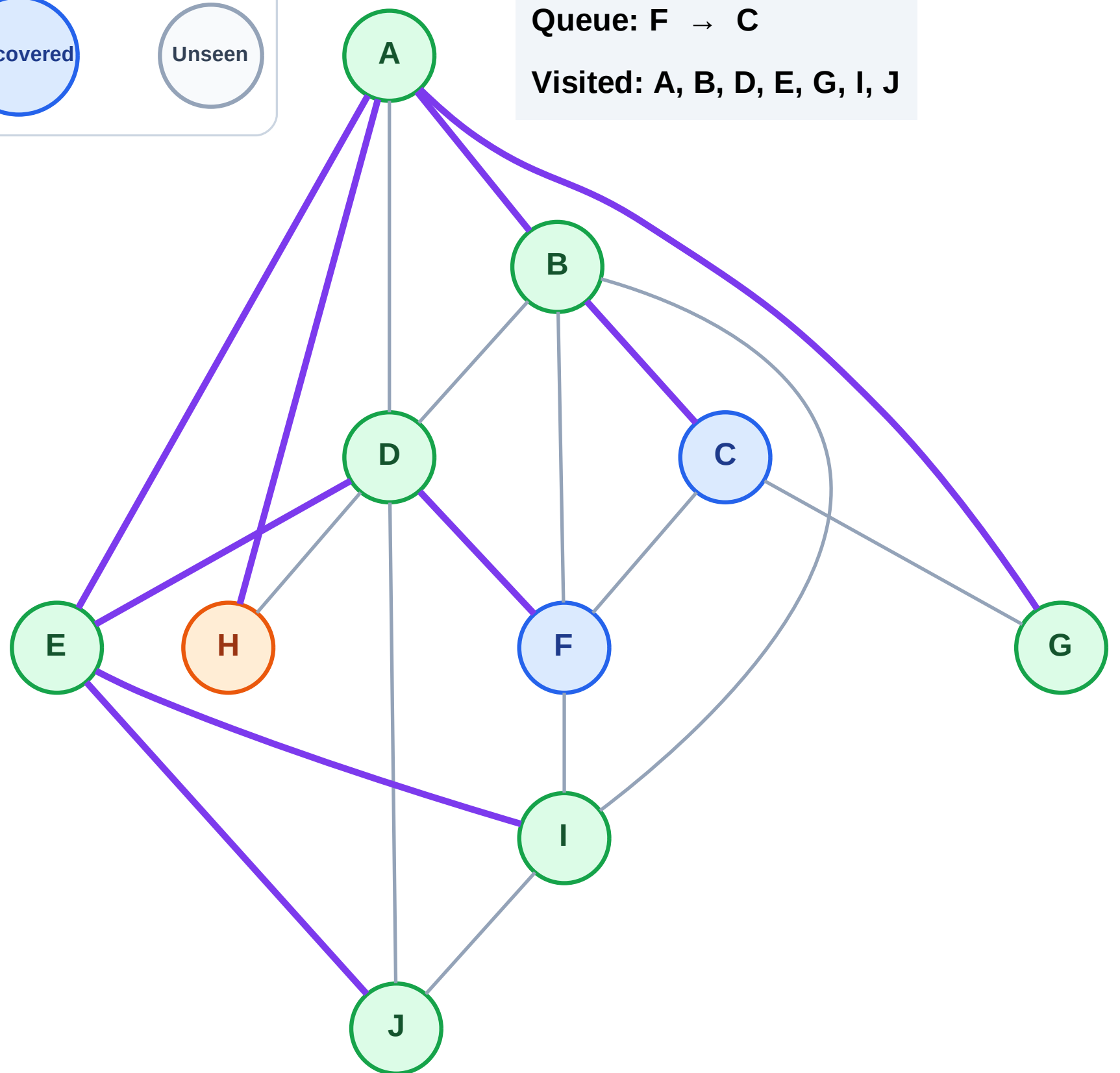
Step 16: Process node H

## Legend



Queue: F → C

Visited: A, B, D, E, G, I, J

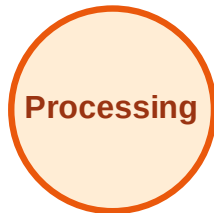




# BFS Traversal

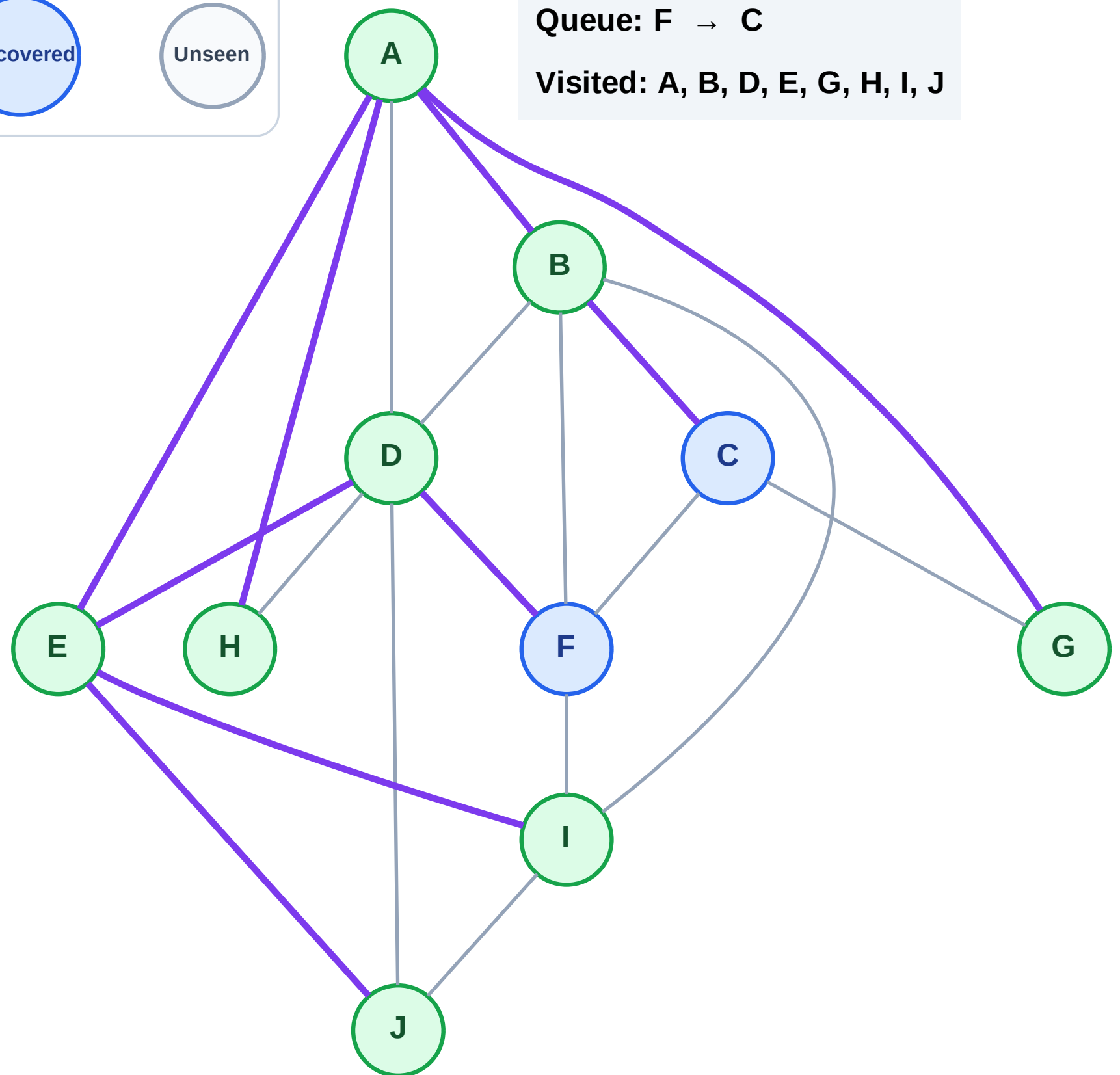
Step 17: No new neighbors from H

## Legend



Queue: F → C

Visited: A, B, D, E, G, H, I, J



# BFS Traversal

Step 18: Process node F

Legend

Complete

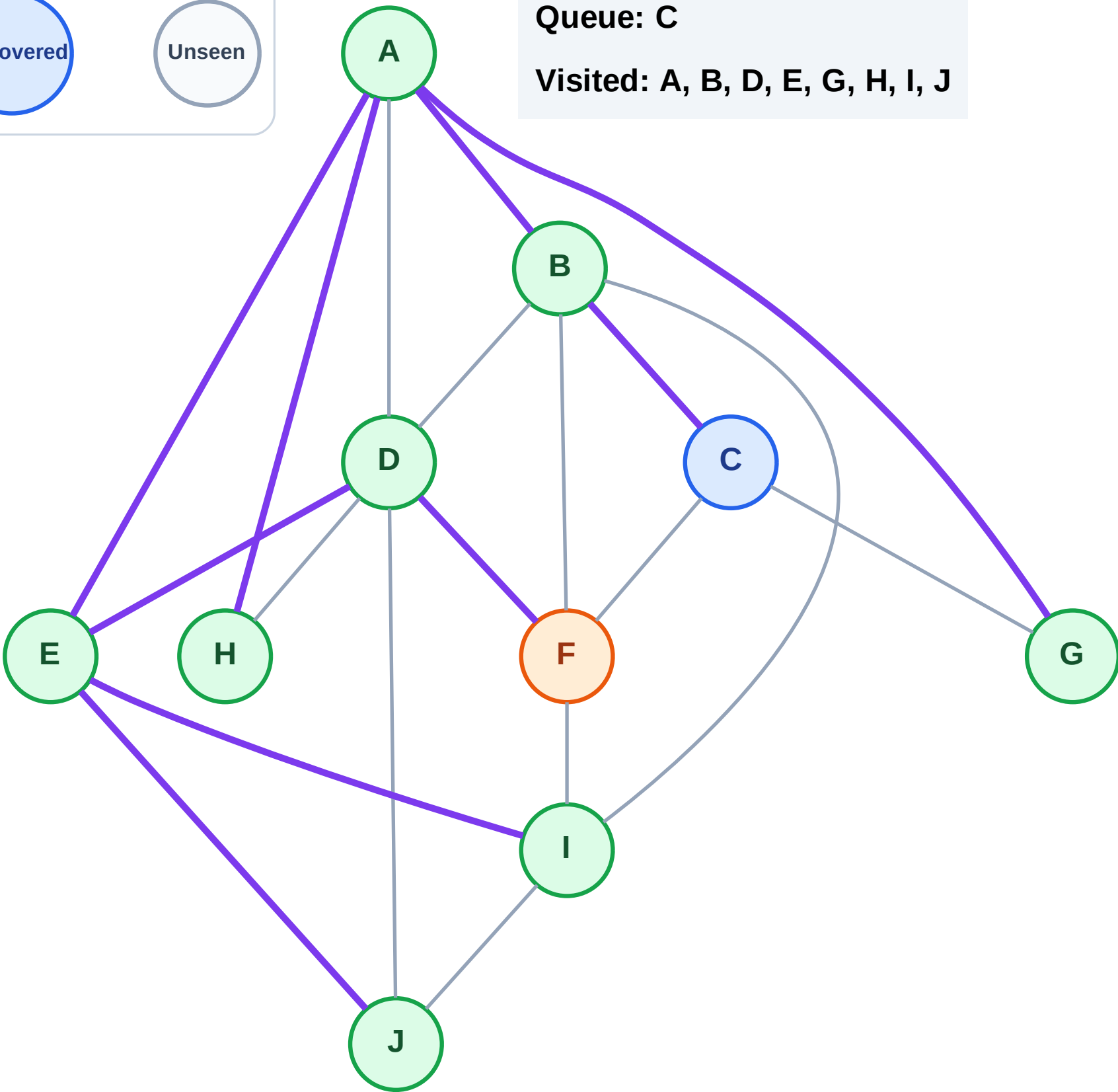
Processing

Discovered

Unseen

Queue: C

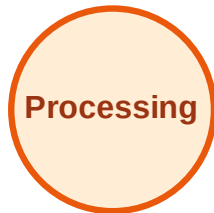
Visited: A, B, D, E, G, H, I, J



# BFS Traversal

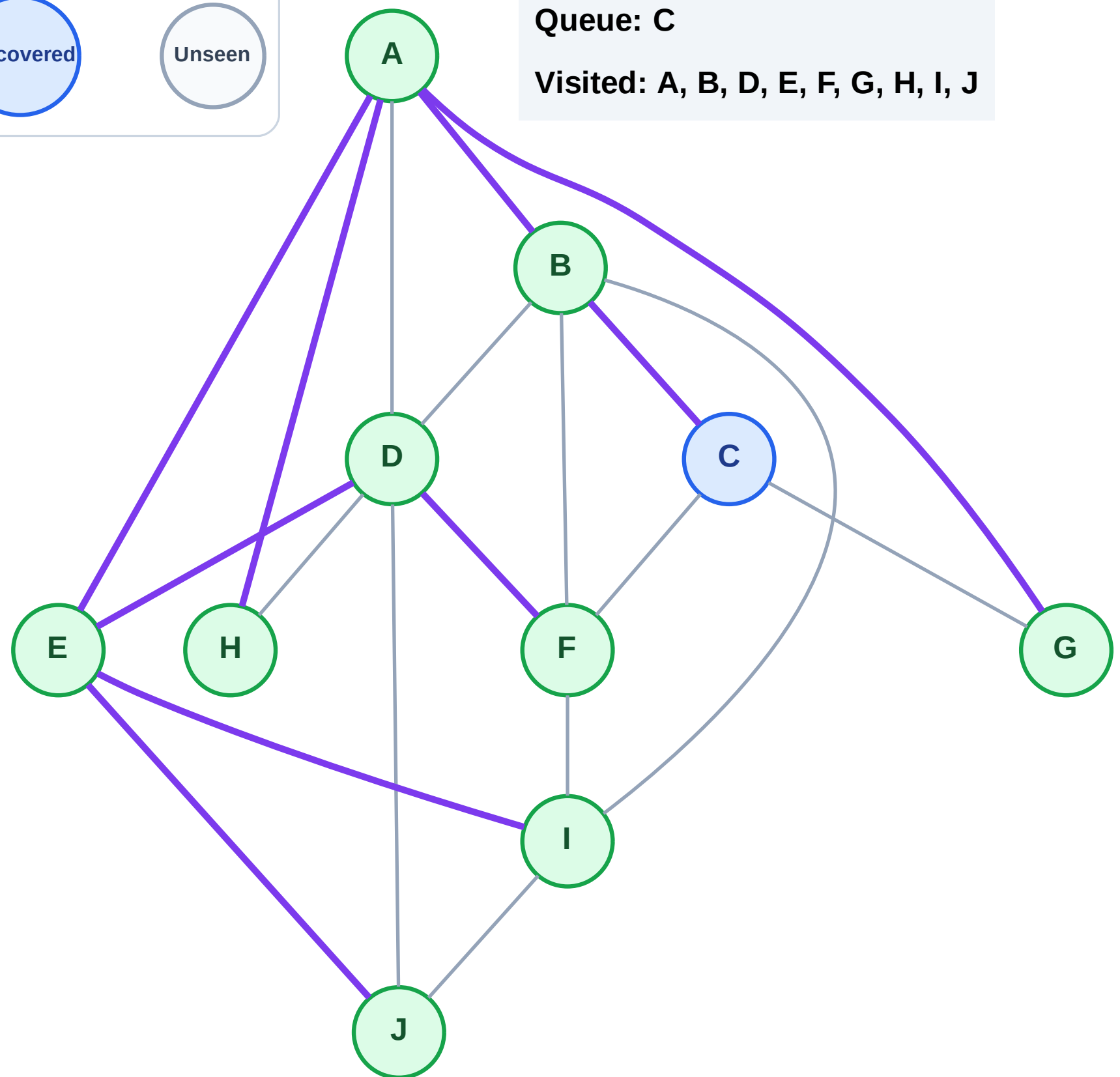
Step 19: No new neighbors from F

## Legend



Queue: C

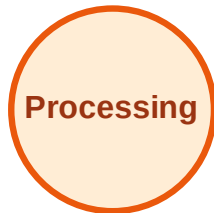
Visited: A, B, D, E, F, G, H, I, J



# BFS Traversal

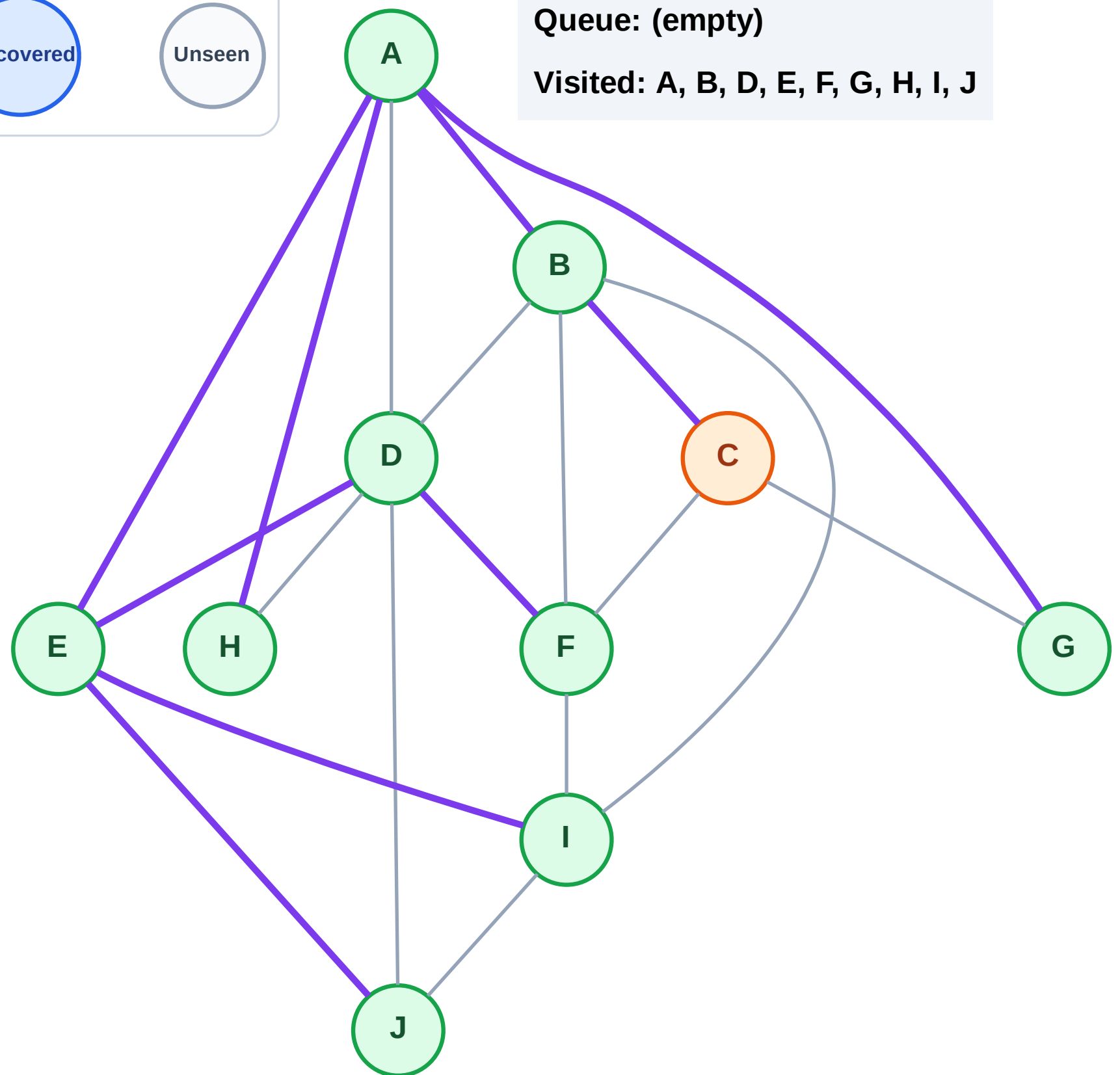
Step 20: Process node C

## Legend



Queue: (empty)

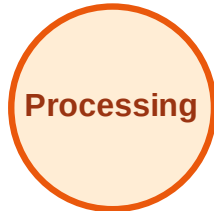
Visited: A, B, D, E, F, G, H, I, J



# BFS Traversal

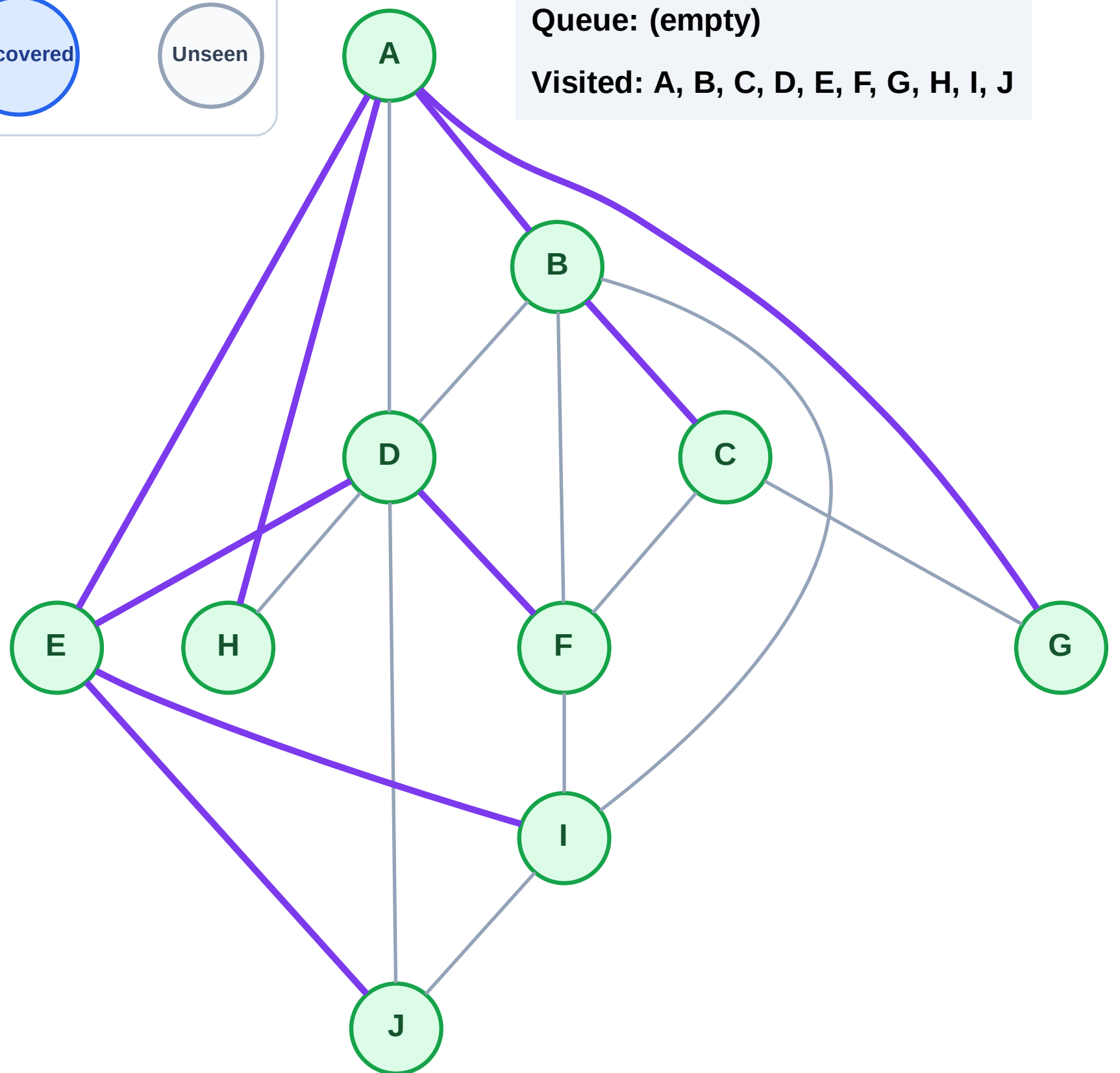
Step 21: No new neighbors from C

## Legend



Queue: (empty)

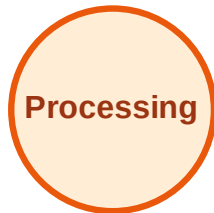
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

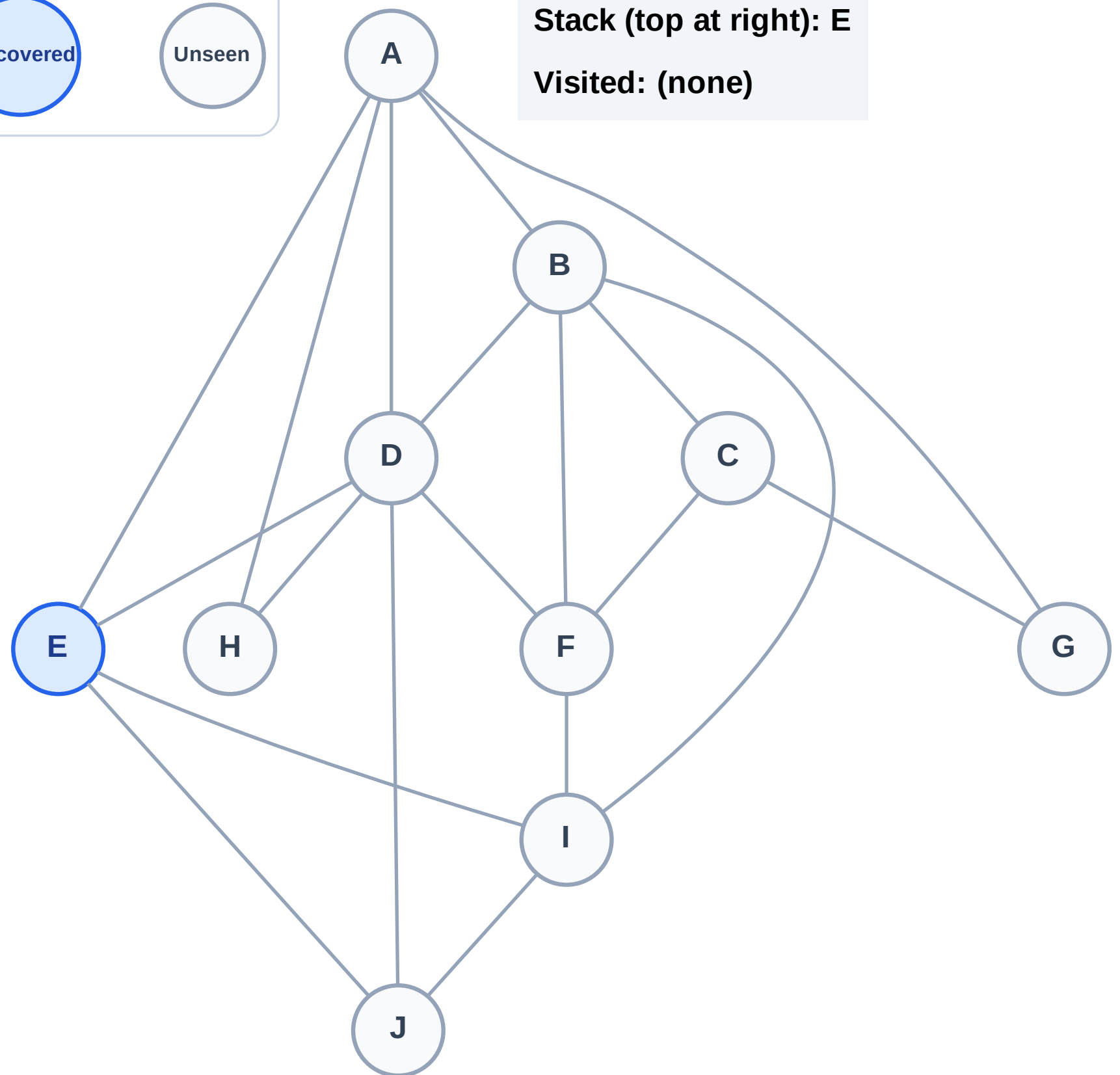
Step 1: Start at E

## Legend



Stack (top at right): E

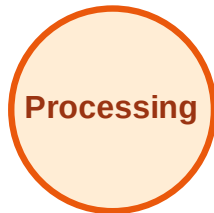
Visited: (none)



# DFS Traversal

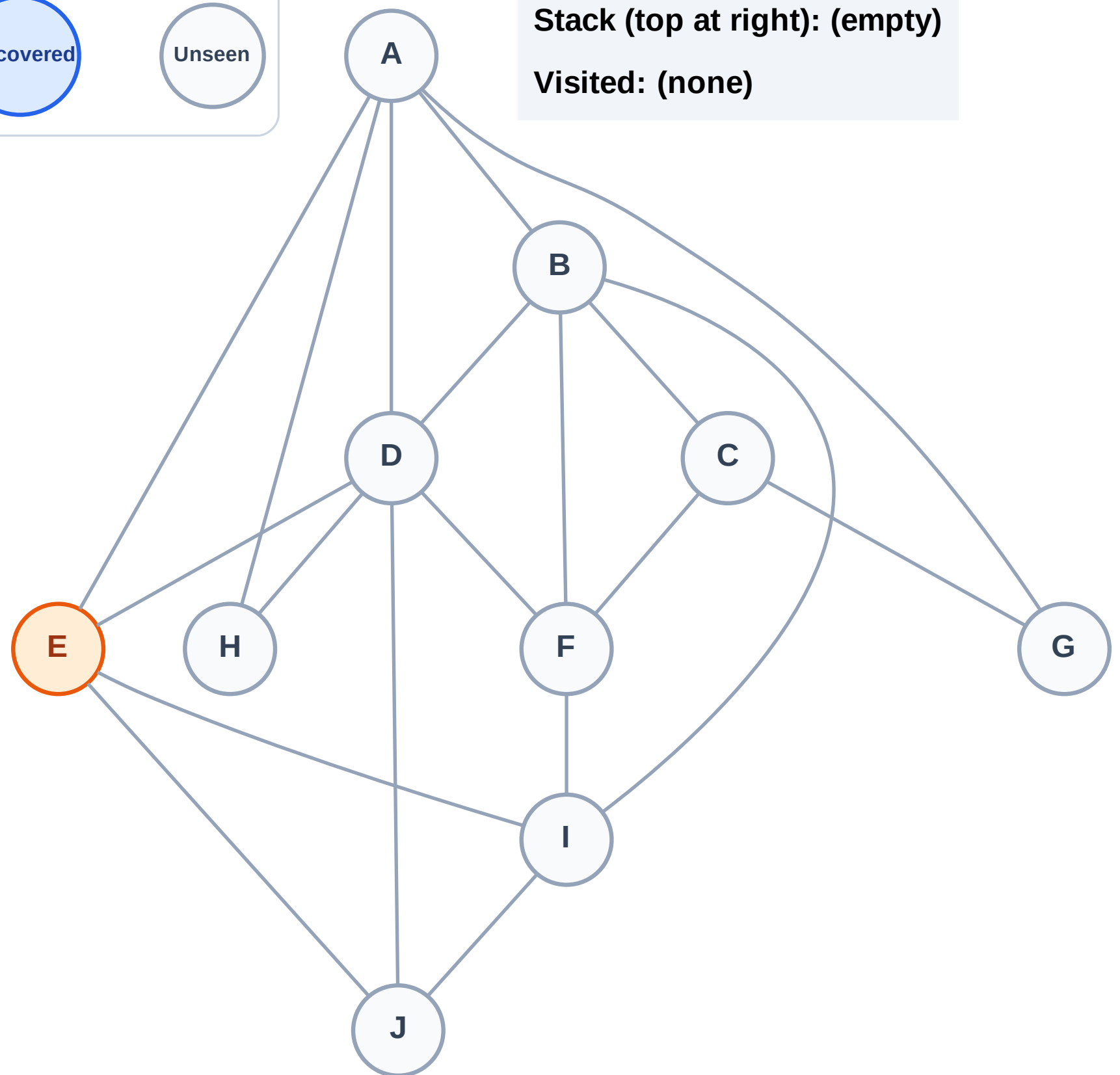
Step 2: Process node E

## Legend



Stack (top at right): (empty)

Visited: (none)

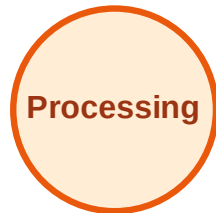




# DFS Traversal

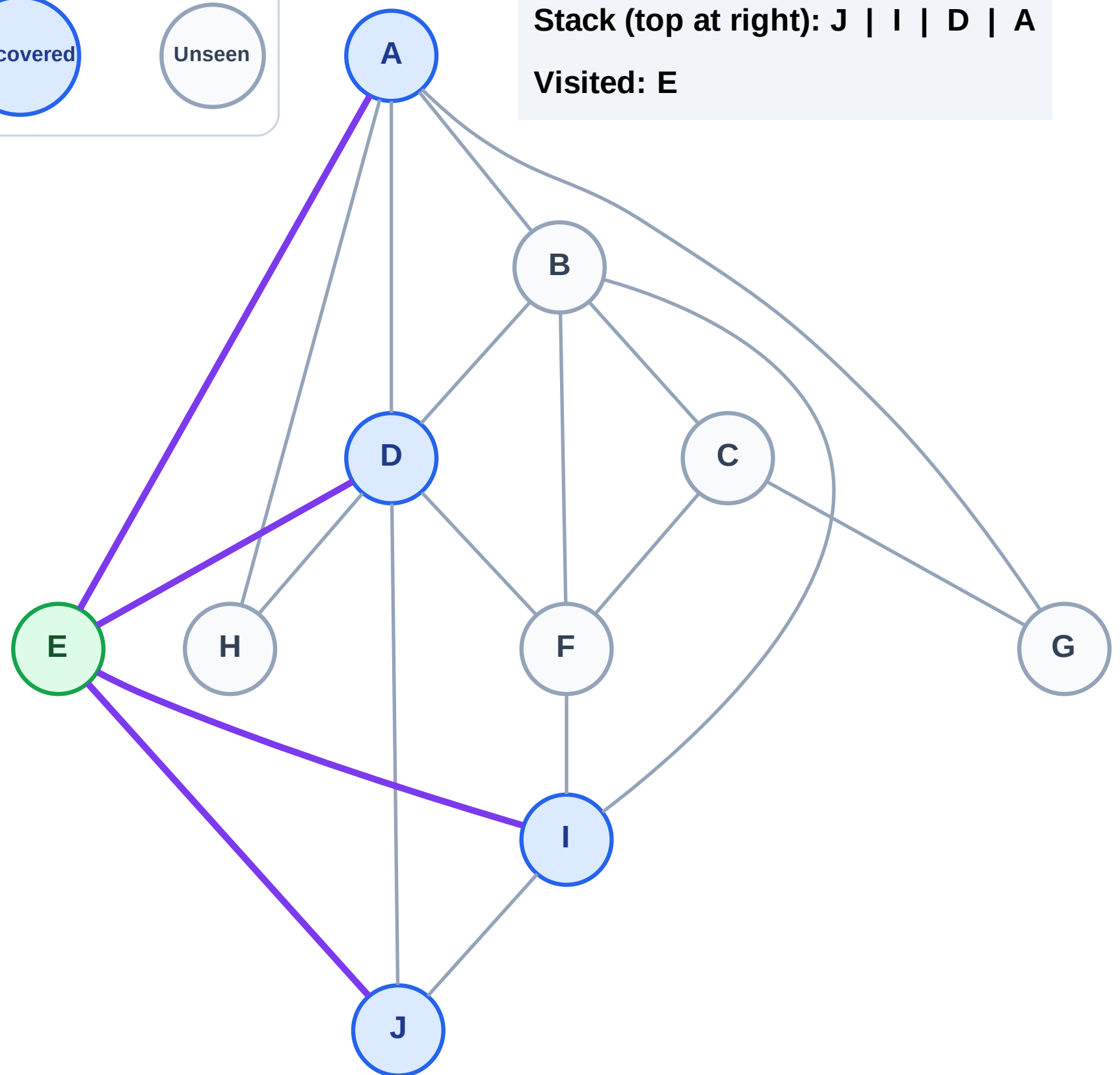
Step 3: Discover J, I, D, A from E

## Legend



Stack (top at right): J | I | D | A

Visited: E

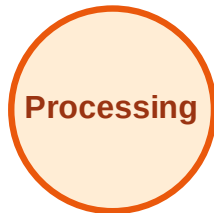




# DFS Traversal

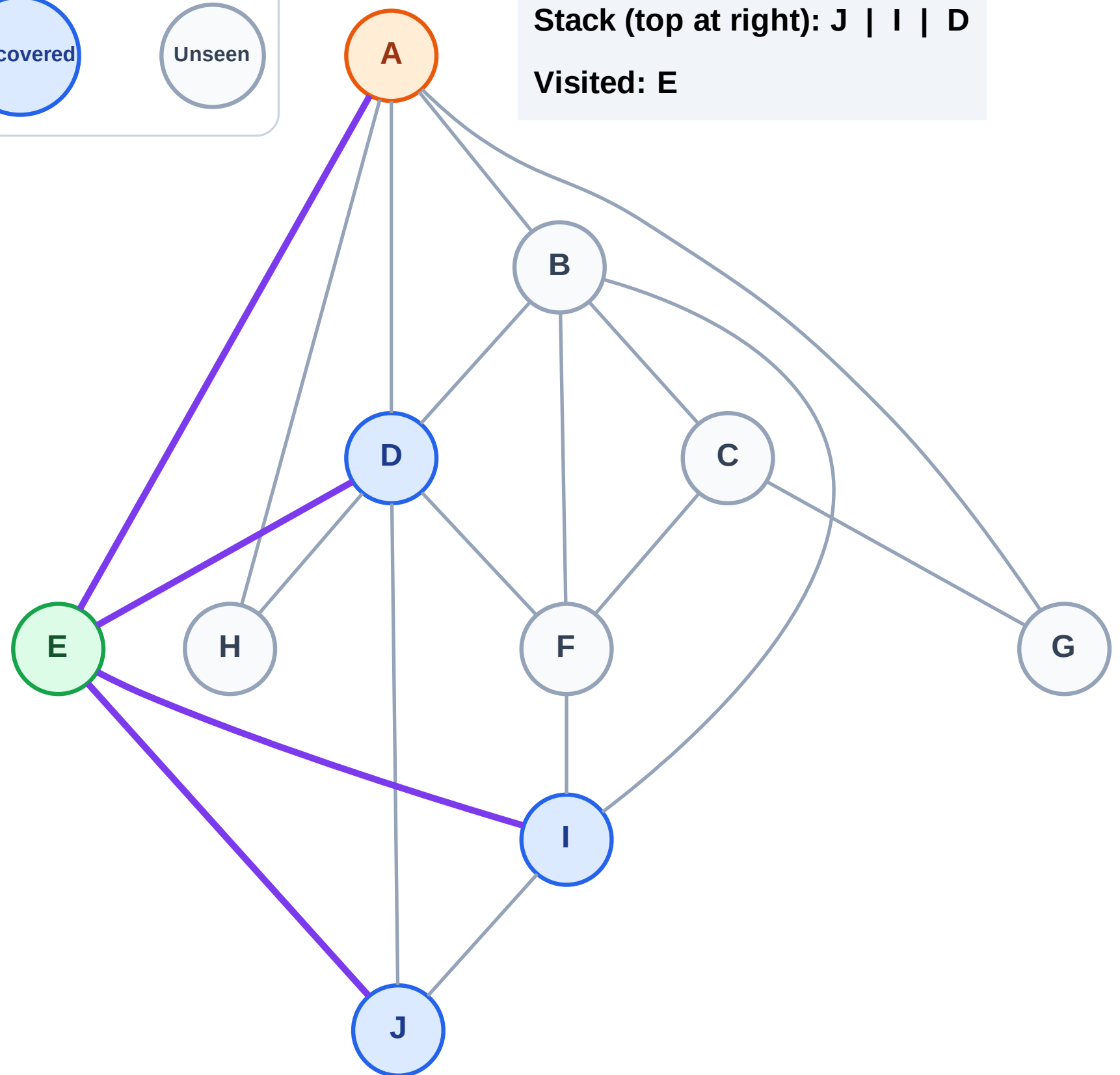
Step 4: Process node A

## Legend



Stack (top at right): J | I | D

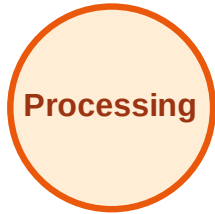
Visited: E



# DFS Traversal

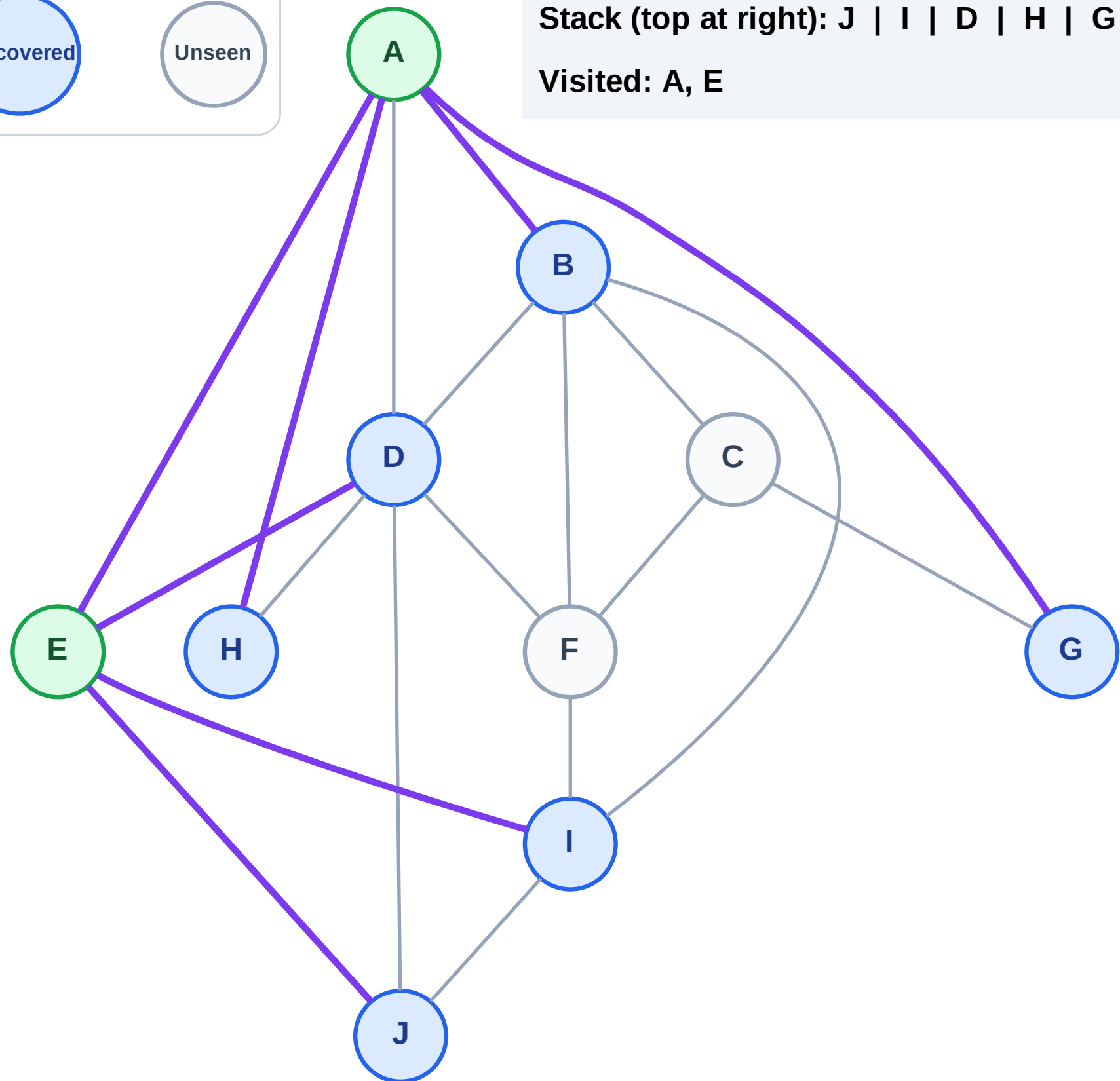
Step 5: Discover H, G, B from A

## Legend



Stack (top at right): J | I | D | H | G | B

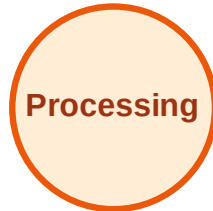
Visited: A, E



# DFS Traversal

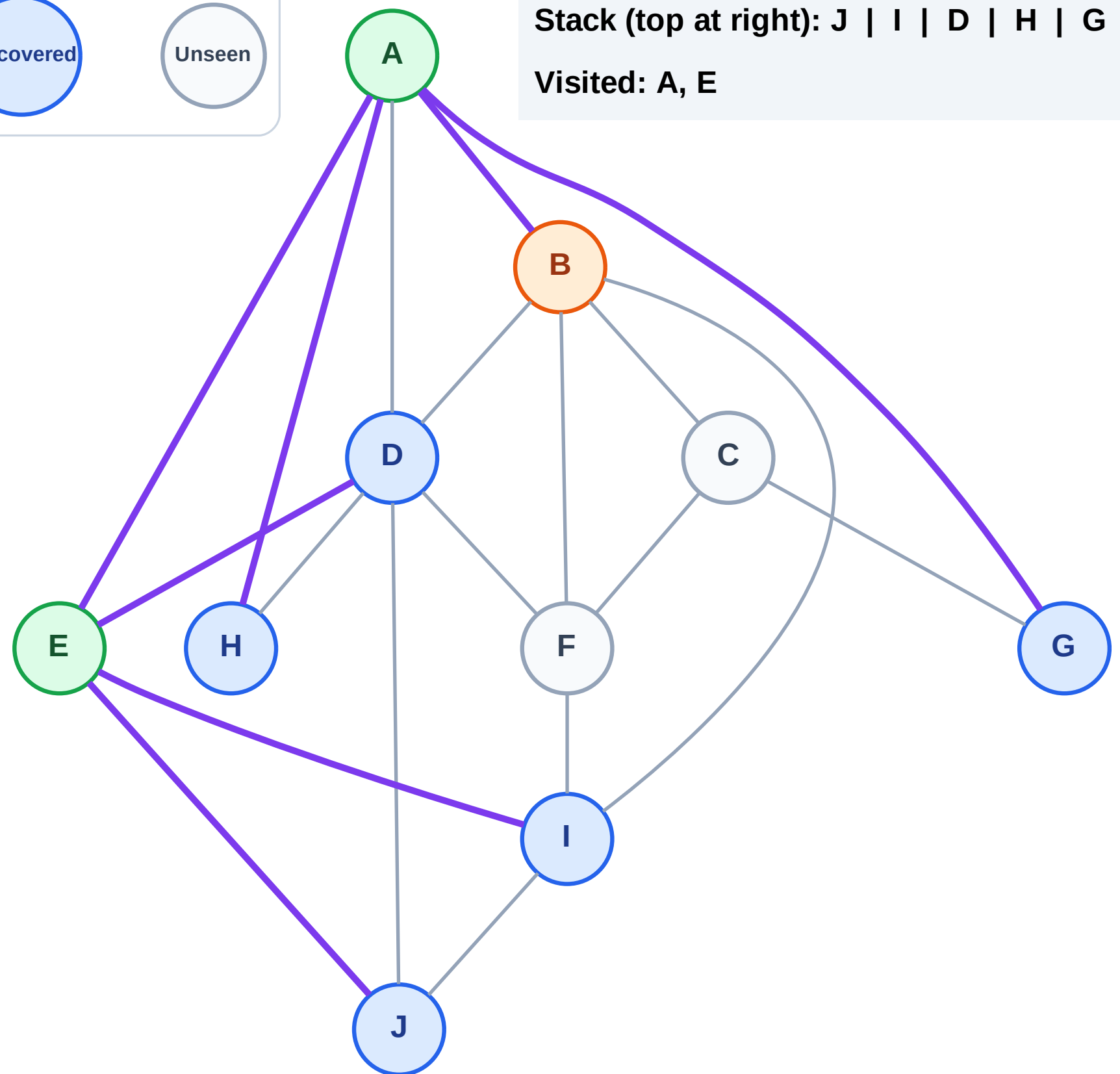
Step 6: Process node B

## Legend



Stack (top at right): J | I | D | H | G

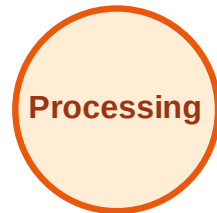
Visited: A, E



# DFS Traversal

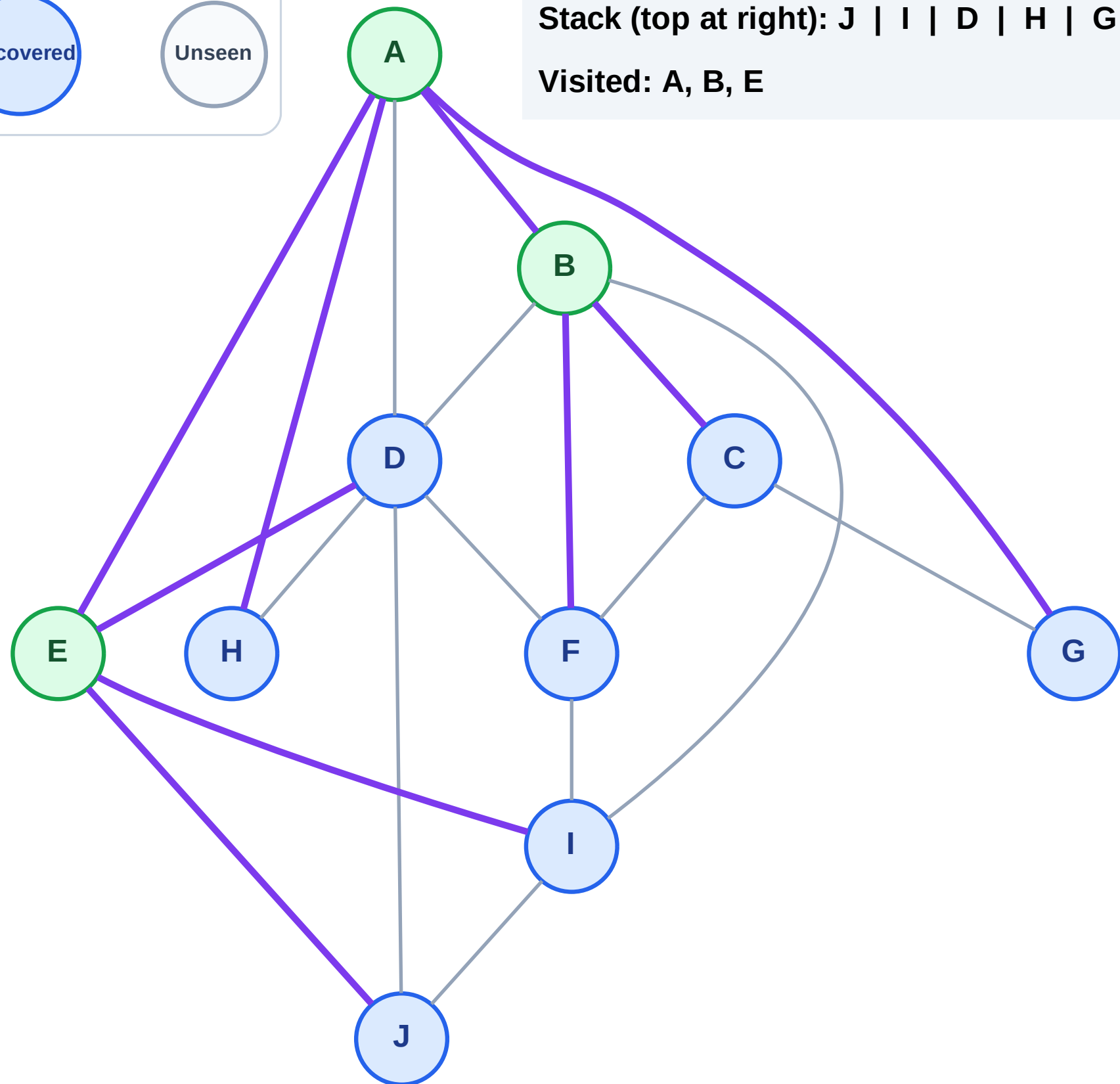
Step 7: Discover F, C from B

## Legend



Stack (top at right): J | I | D | H | G | F | C

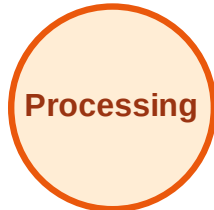
Visited: A, B, E



# DFS Traversal

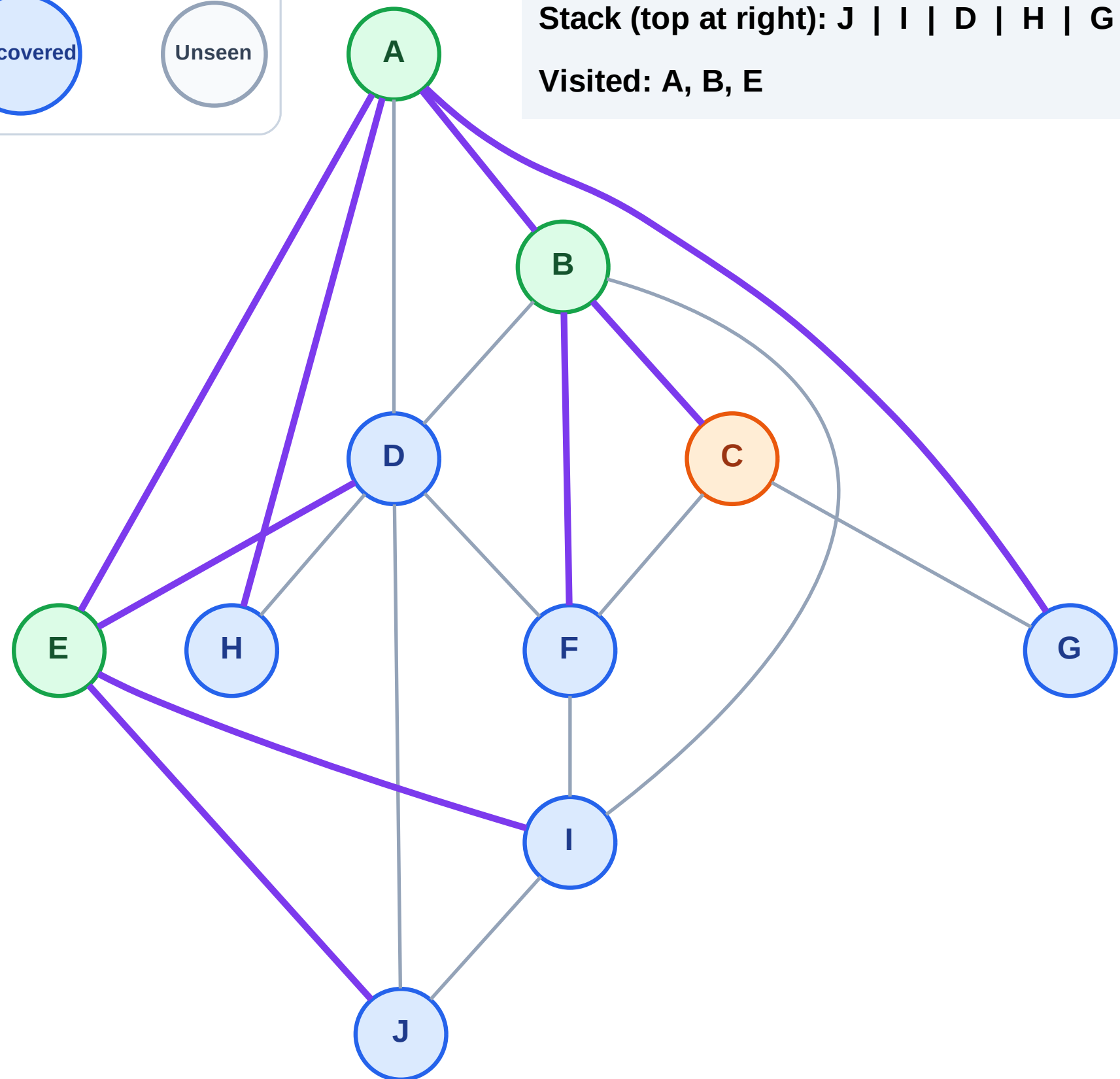
Step 8: Process node C

## Legend



Stack (top at right): J | I | D | H | G | F

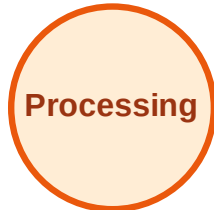
Visited: A, B, E



# DFS Traversal

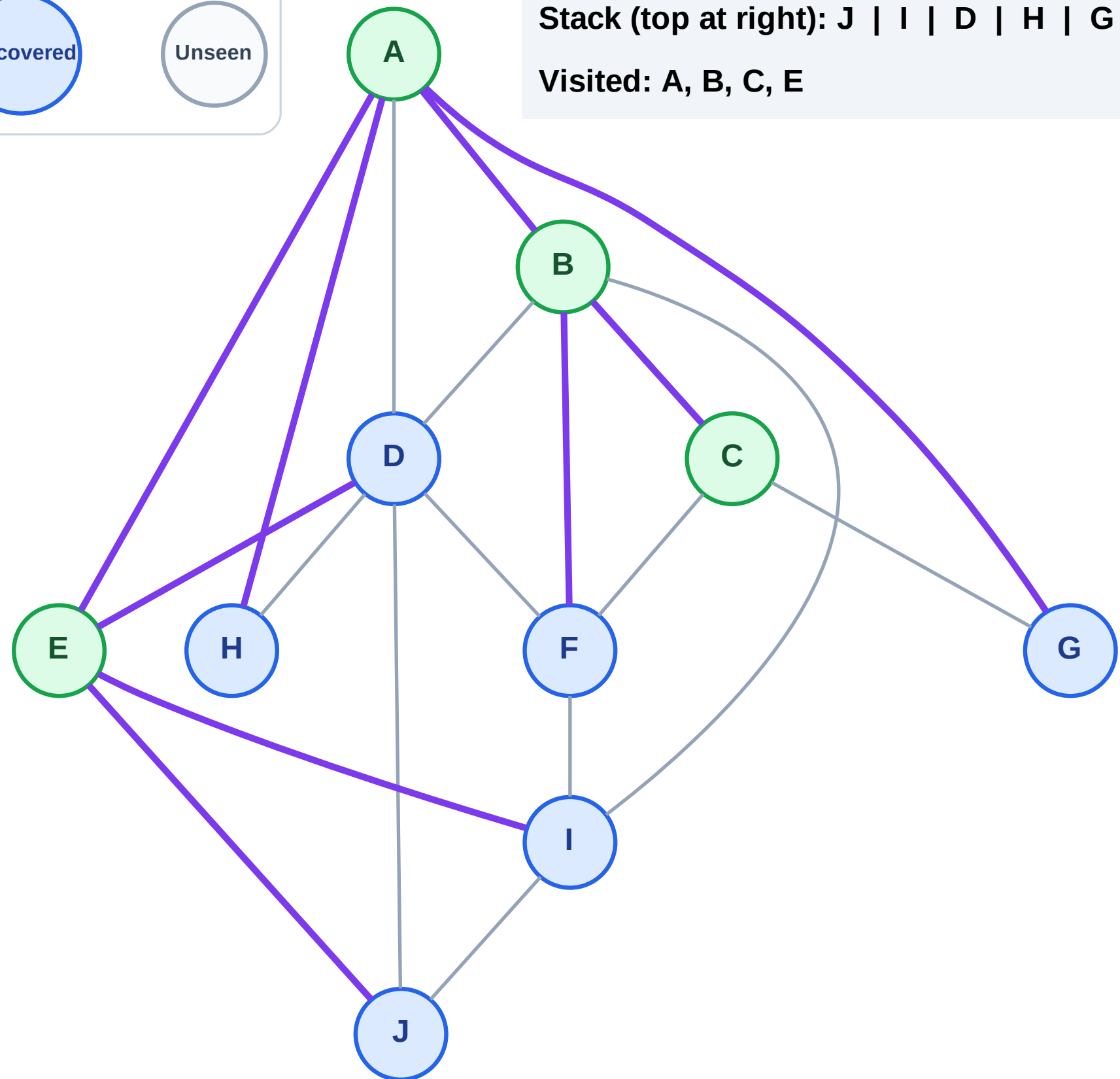
Step 9: No new neighbors from C

## Legend



Stack (top at right): J | I | D | H | G | F

Visited: A, B, C, E



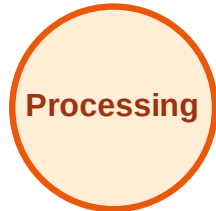
# DFS Traversal

Step 10: Process node F

## Legend



Complete



Processing



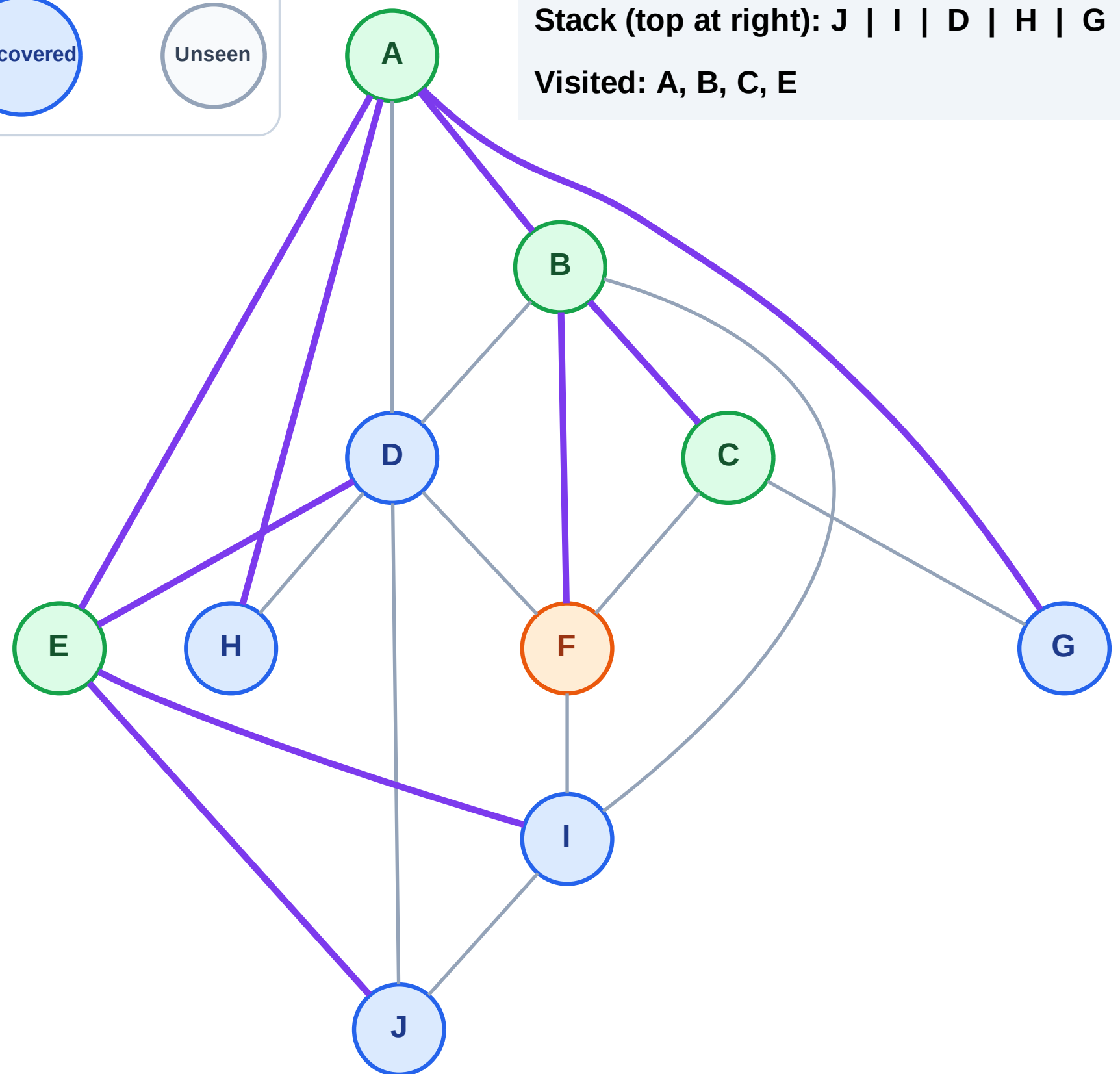
Discovered



Unseen

Stack (top at right): J | I | D | H | G

Visited: A, B, C, E





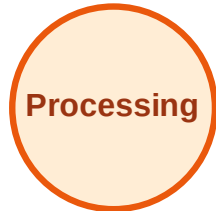
# DFS Traversal

Step 11: No new neighbors from F

## Legend



Complete



Processing



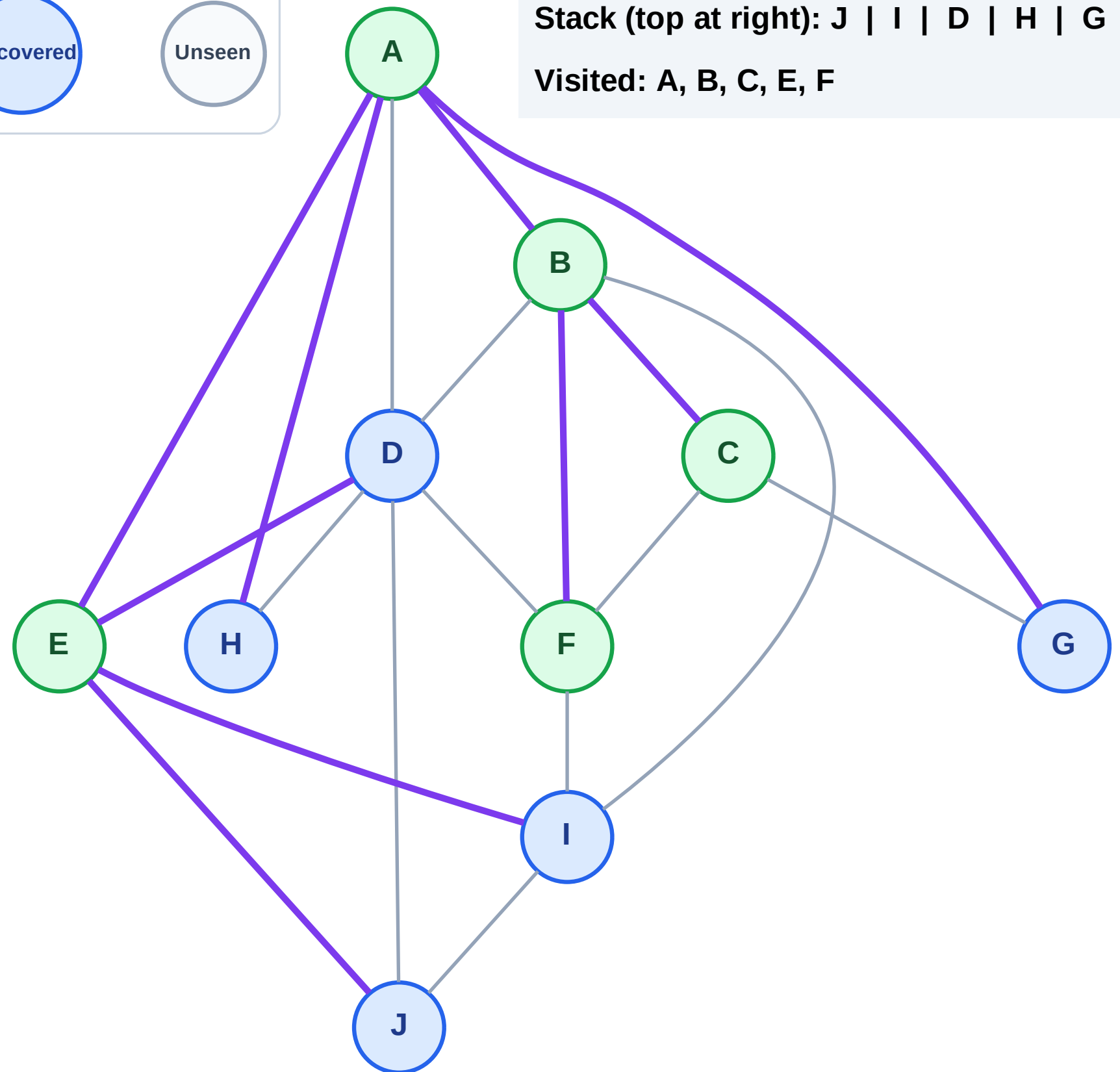
Discovered



Unseen

Stack (top at right): J | I | D | H | G

Visited: A, B, C, E, F

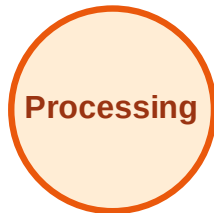




# DFS Traversal

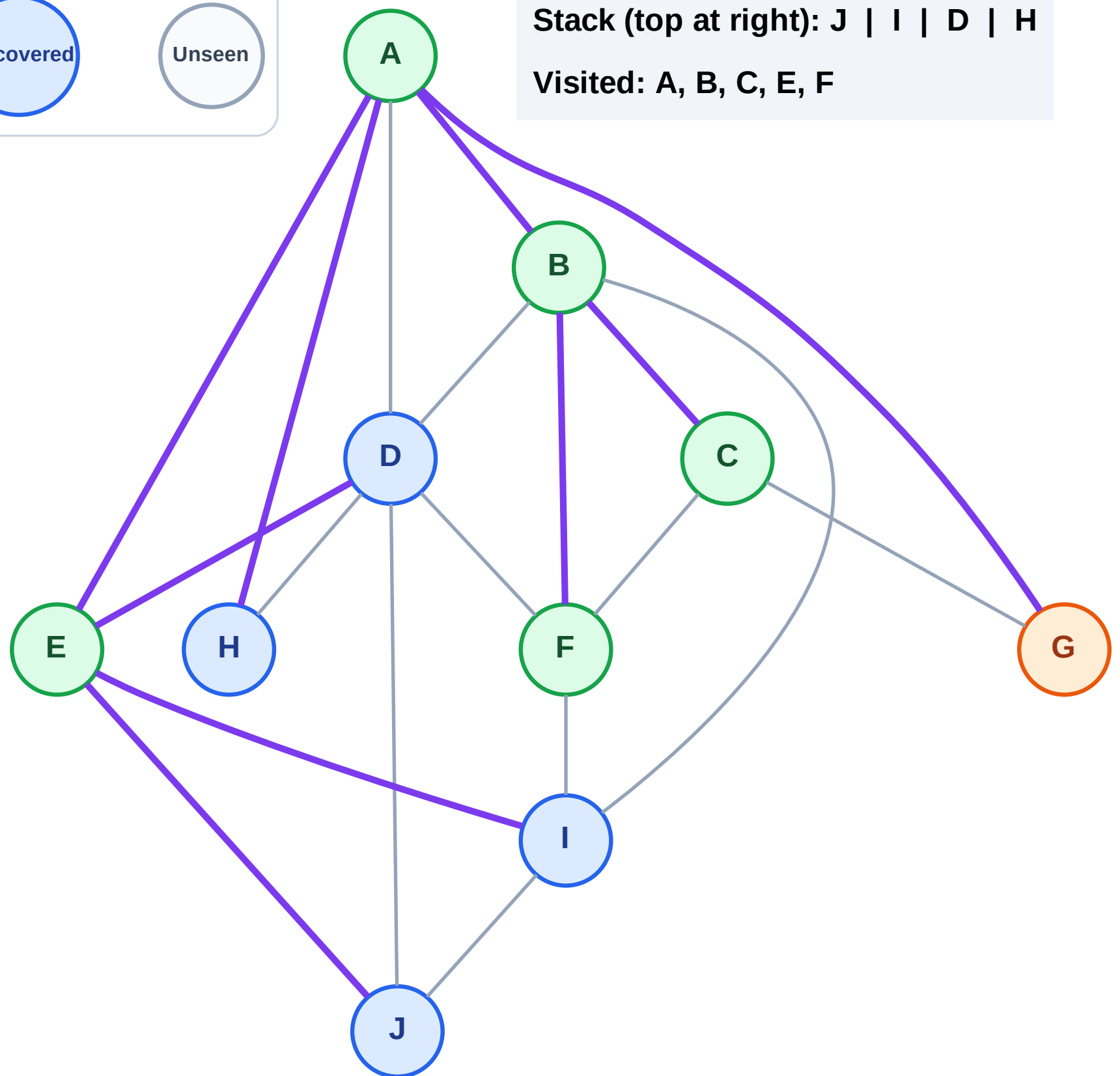
Step 12: Process node G

## Legend



Stack (top at right): J | I | D | H

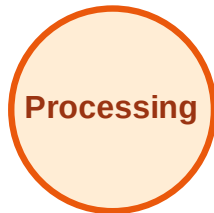
Visited: A, B, C, E, F



# DFS Traversal

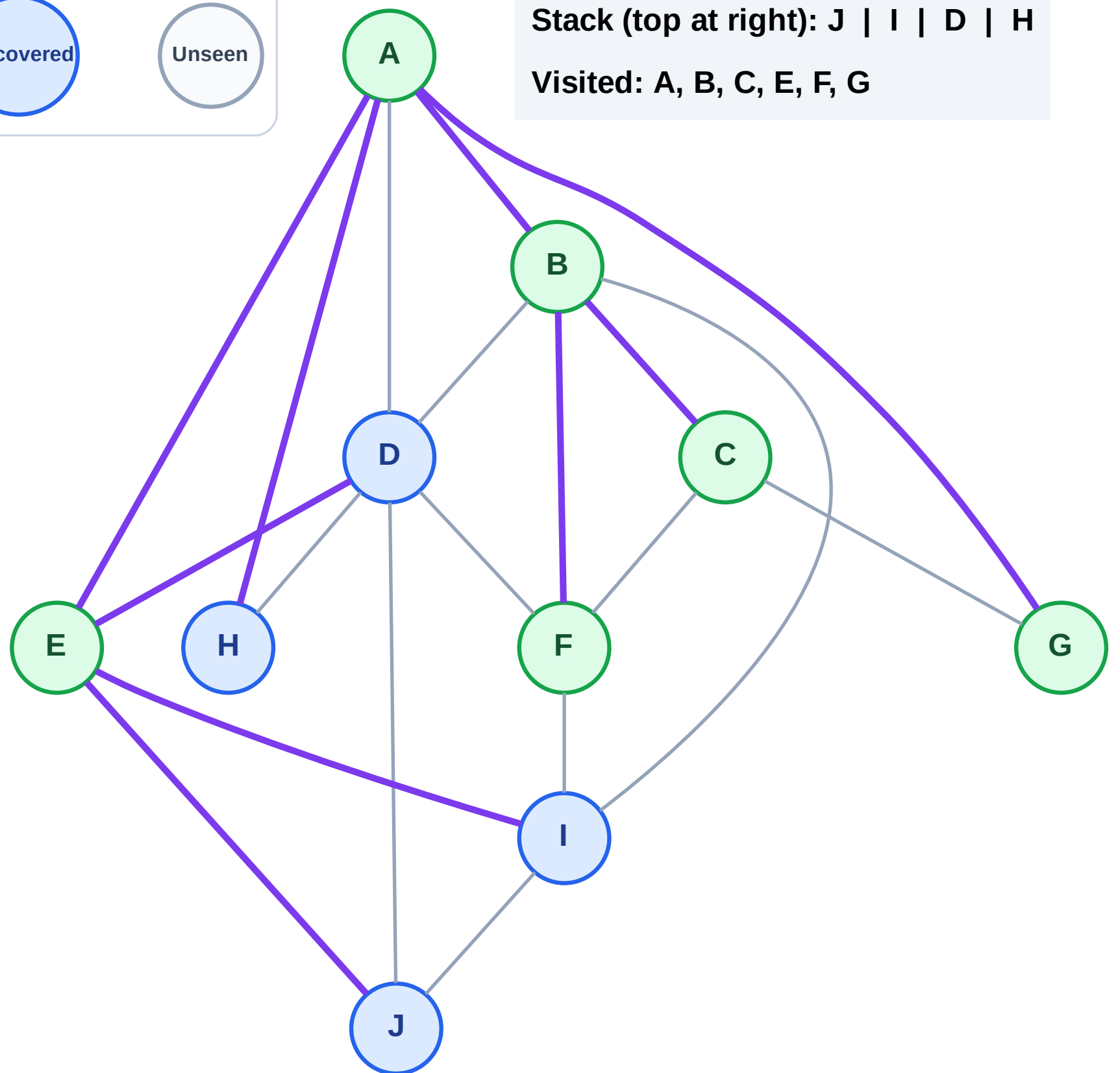
Step 13: No new neighbors from G

## Legend



Stack (top at right): J | I | D | H

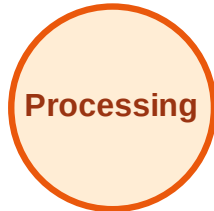
Visited: A, B, C, E, F, G



# DFS Traversal

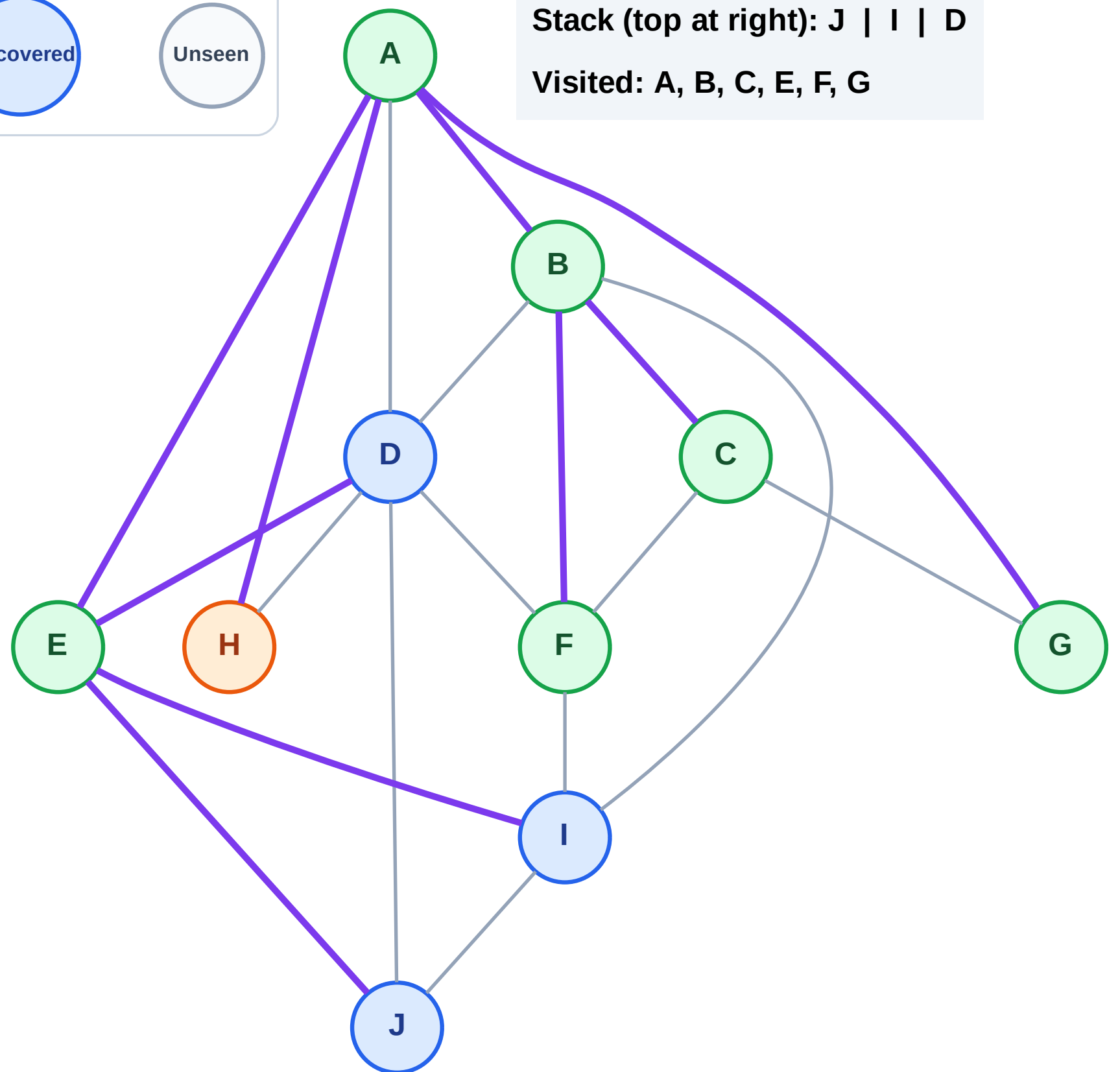
Step 14: Process node H

## Legend



Stack (top at right): J | I | D

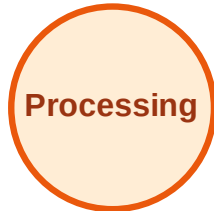
Visited: A, B, C, E, F, G



# DFS Traversal

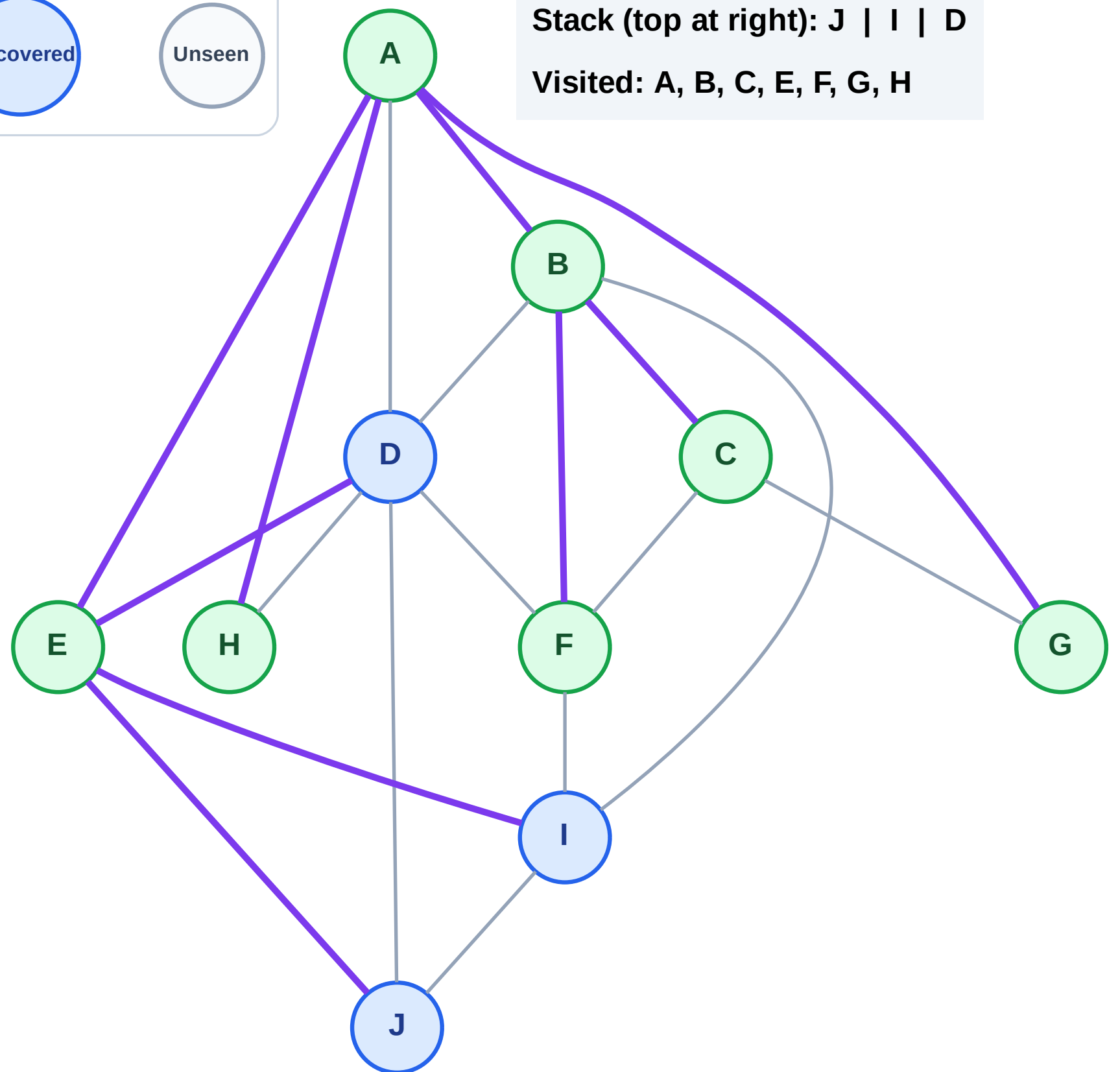
Step 15: No new neighbors from H

## Legend



Stack (top at right): J | I | D

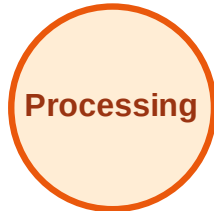
Visited: A, B, C, E, F, G, H



# DFS Traversal

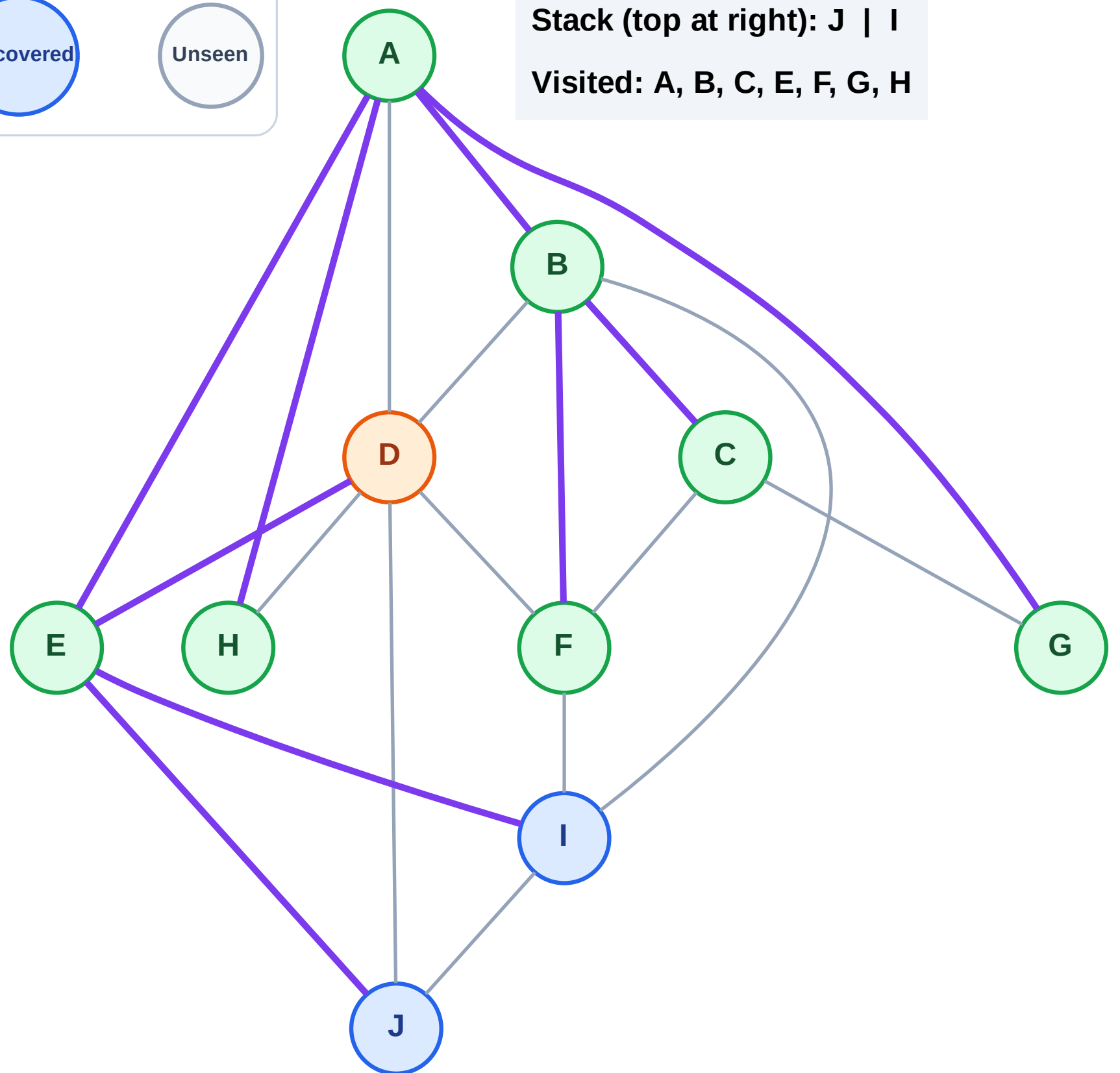
Step 16: Process node D

## Legend



Stack (top at right): J | I

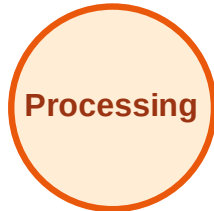
Visited: A, B, C, E, F, G, H



# DFS Traversal

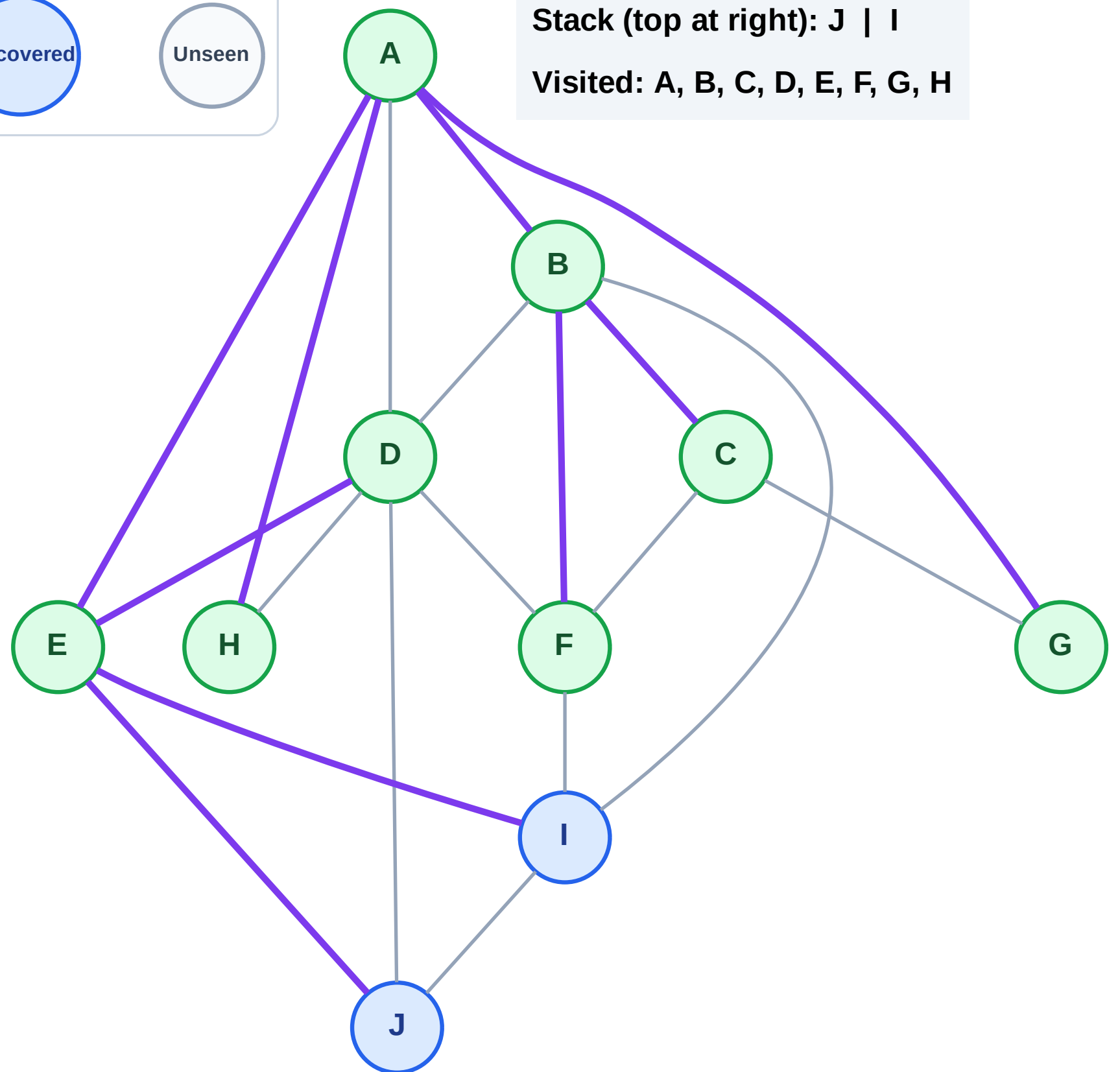
Step 17: No new neighbors from D

## Legend



Stack (top at right): J | I

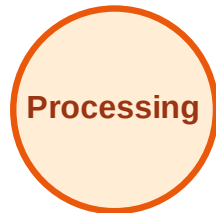
Visited: A, B, C, D, E, F, G, H



# DFS Traversal

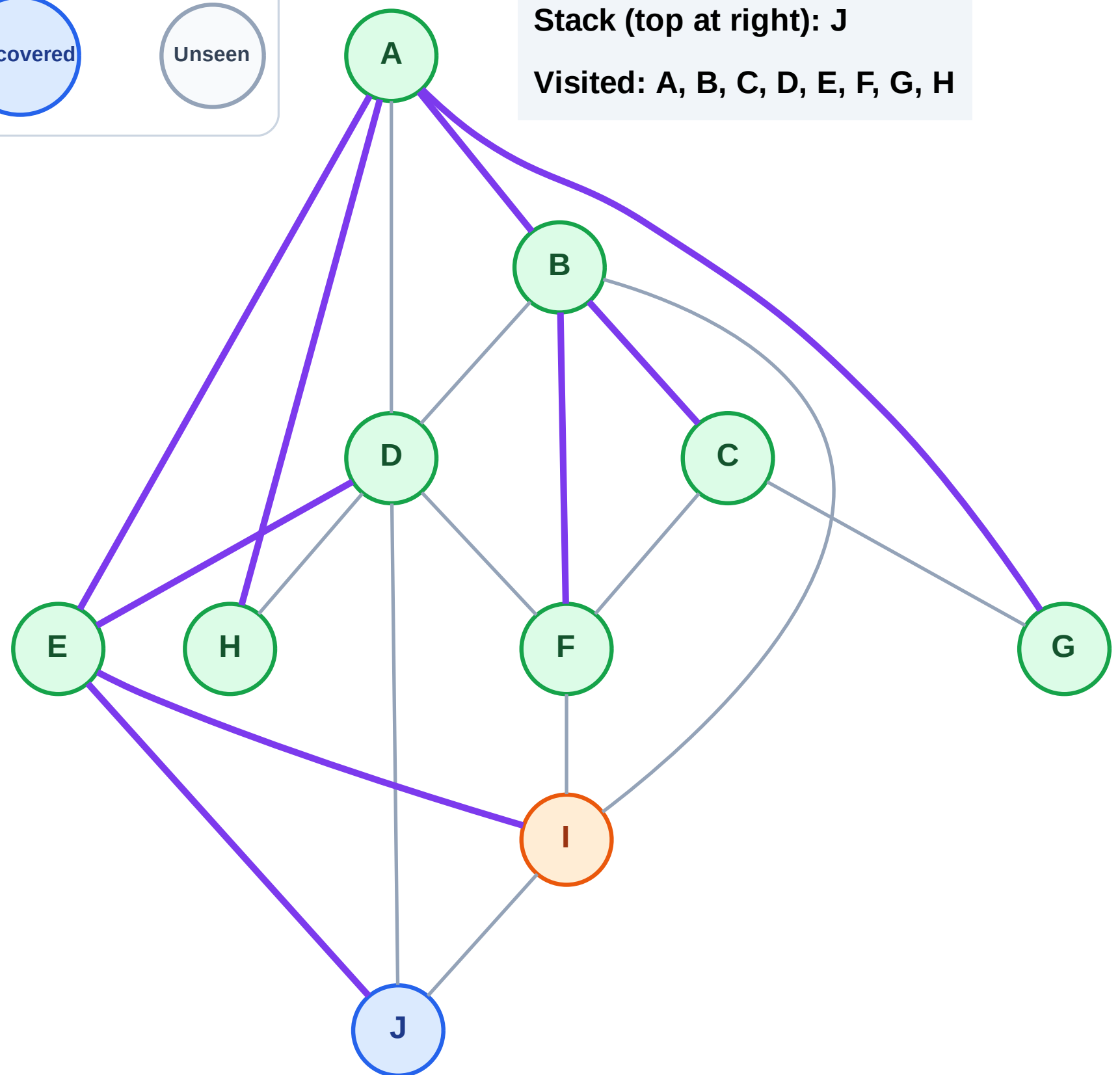
Step 18: Process node I

## Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H

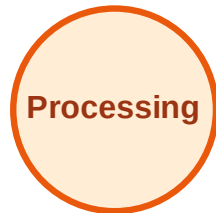




# DFS Traversal

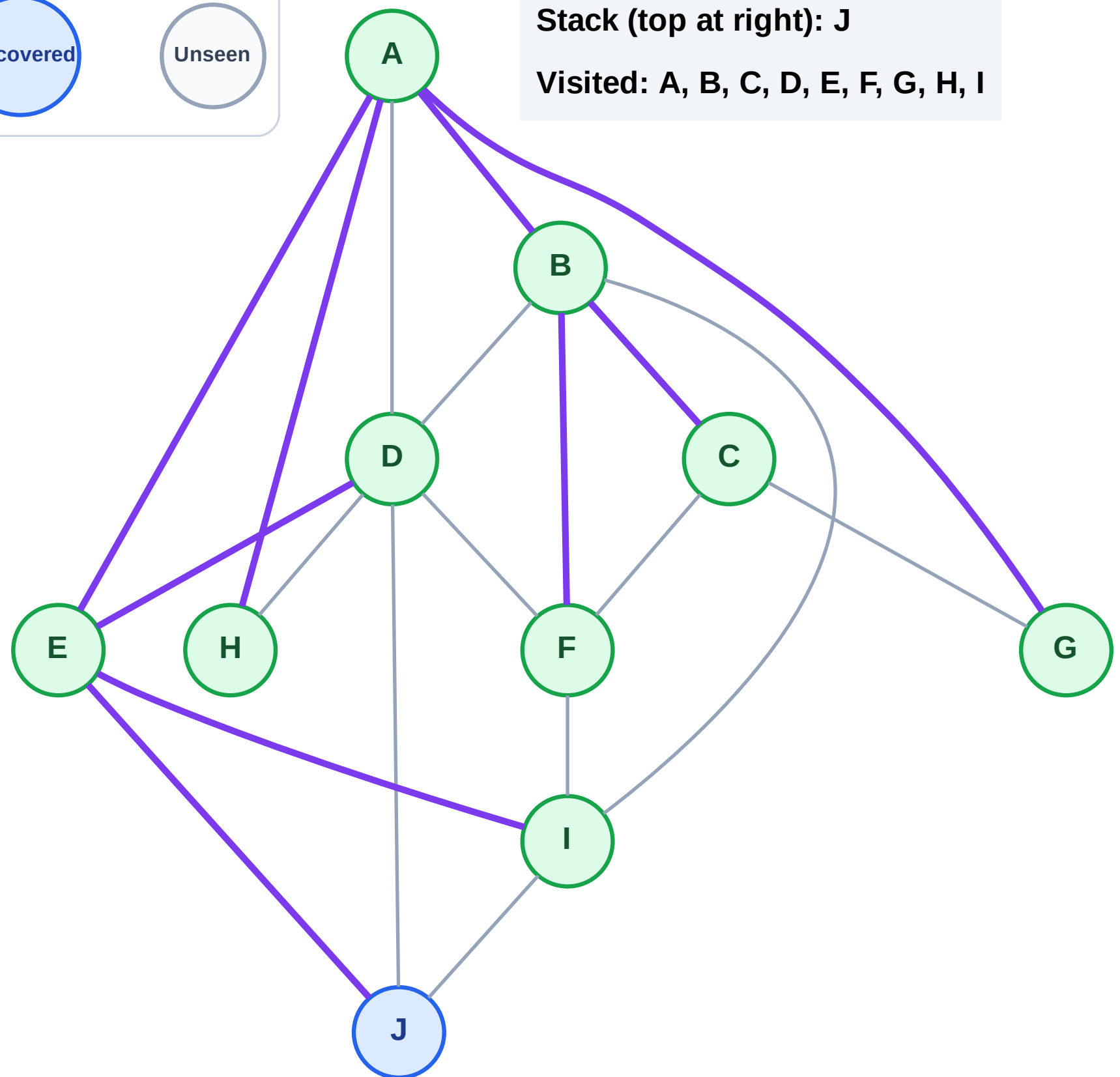
Step 19: No new neighbors from I

## Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I

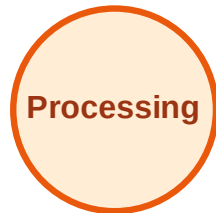




# DFS Traversal

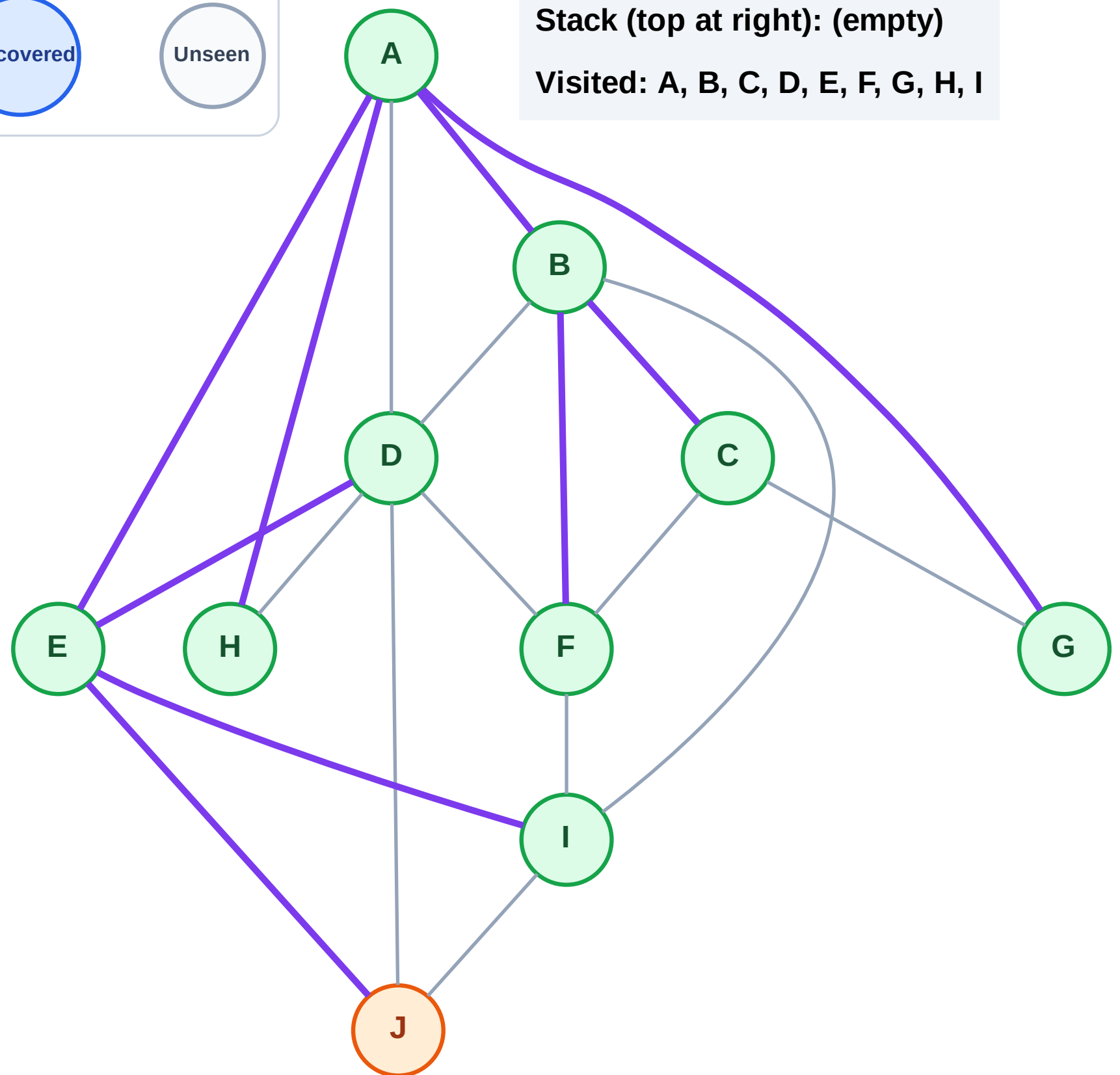
Step 20: Process node J

## Legend



Stack (top at right): (empty)

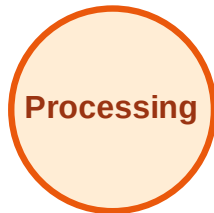
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

